ESE 3105 Complete Solutions

Heat and Mass Transfer · KUET · past papers 2019, 2020, 2021, 2022, 2023 · answers from the course notes (Mahmudul sir, Mahfuz notes, Sohag sir) and the Cengel *Heat and Mass Transfer* textbook they follow

24-hour priority list

You answer **3 of 4 questions in each part**, so each part has its own list. Study down both lists until the green **pass line**: about 3 h for Part A and 4 h for Part B, which leaves time for sleep and a last read of the one-shot notes. Alternate between parts (A block 1, B block 1, A block 2 …) so neither is left short. **Chance**: how often the topic came up, newer papers weigh more (2023 ×5, 2022 ×4, 2021 ×3, 2020 ×2, 2019 ×1). **Covered**: share of that part's past-paper marks you have studied once you finish the row.

Retake tip: read the Concept Primer first (30 minutes). In Part A the wall/pipe resistance numerical (A2), critical radius (A3) and the fin derivation plus numerical (A4) come every year and all have calculators. The boiling curve (A6) is pure theory: learn the five regions. In Part B learn the dimensionless numbers (B1) and the flat-plate numerical (B3); natural convection on a plate (B6) is the easiest numerical. Tube banks (B5) are long: do them last.

Section A: Conduction, insulation, fins, transient, boiling and condensation

Block 1: very likely and easy · 1.2 h

Topic	Chance	Level	Covered
A6 Boiling and condensation	Very likely 100%	Easy	26%
A2 Steady one-dimensional conduction: walls, cylinders, spheres, thermal resistance, heat generation	Very likely 100%	Easy	44%
A1 Conduction basics: modes, k and α, boundary conditions, general conduction equation	Very likely 87%	Easy	59%

Block 2: very likely, medium effort · 2.2 h

Topic	Chance	Level	Covered
A4 Fins (extended surfaces): equation, solutions, efficiency, effectiveness, numericals	Very likely 100%	Medium	74%
A3 Critical radius (thickness) of insulation	Very likely 100%	Medium	82%
✔ Pass line: 82% of Section A marks covered, enough to choose 3 of the 4 questions (about 2.8 h)			
A5 Transient conduction: lumped system, Heisler/one-term solution, semi-infinite solid	Very likely 87%	Medium	100%

Section B: Convection and mass transfer

Block 1: very likely and easy · 0.4 h

Topic	Chance	Level	Covered
B1 Convection fundamentals: mechanism, boundary layers, dimensionless numbers, film temperature	Very likely 100%	Easy	22%

Block 2: very likely, medium effort · 1.5 h

Topic	Chance	Level	Covered
B7 Mass transfer: Fick's law, diffusion, analogies, numericals	Very likely 100%	Medium	43%
B2 Boundary-layer analysis: integral momentum and energy equations, Nu from profiles, Reynolds-Colburn analogy	Very likely 80%	Medium	60%

Block 3: very likely numericals · 1.0 h

Topic	Chance	Level	Covered
B5 Flow across tube banks	Very likely 87%	Hard	73%

Block 4: likely · 1.5 h

Topic	Chance	Level	Covered
B6 Natural (free) convection	Likely 67%	Medium	81%
✔ Pass line: 81% of Section B marks covered, enough to choose 3 of the 4 questions (about 3.7 h)			
B4 Internal forced convection: fully developed flow, entry lengths, analogy, pipe numericals	Likely 60%	Medium	93%

Only if time is left · 0.8 h

Topic	Chance	Level	Covered
B3 External forced convection over a flat plate: numericals	Possible 47%	Medium	100%

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2. **Section A: Conduction, insulation, fins, transient, boiling and condensation**: A6 Boiling and condensation, A2 Steady one-dimensional conduction: walls, cylinders, spheres, thermal resistance, heat generation, A1 Conduction basics: modes, k and α, boundary conditions, general conduction equation, A4 Fins (extended surfaces): equation, solutions, efficiency, effectiveness, numericals, A3 Critical radius (thickness) of insulation, A5 Transient conduction: lumped system, Heisler/one-term solution, semi-infinite solid
3. **Section B: Convection and mass transfer**: B1 Convection fundamentals: mechanism, boundary layers, dimensionless numbers, film temperature, B7 Mass transfer: Fick's law, diffusion, analogies, numericals, B2 Boundary-layer analysis: integral momentum and energy equations, Nu from profiles, Reynolds-

Colburn analogy, B5 Flow across tube banks, B6 Natural (free) convection, B4 Internal forced convection: fully developed flow, entry lengths, analogy, pipe numericals, B3 External forced convection over a flat plate: numericals

One-shot notes: ESE 3105 Heat and Mass Transfer

Modes and conduction equation (A1)

Conduction $Q = -kA\,dT/dx$ (Fourier); convection $Q = hA(T_s - T_\infty)$ (Newton); radiation $Q = \varepsilon\sigma A(T_s^4 - T_{sur}^4)$, $\sigma = 5.67\times10^{-8}$. $\alpha = k/\rho c_p$ (how fast heat spreads).

Cartesian: $\partial/\partial x(k\partial T/\partial x) + \ldots + \dot{e} = \rho c\,\partial T/\partial t$.

Cylinder: $(1/r)\partial/\partial r(kr\partial T/\partial r) + (1/r^2)\partial/\partial\varphi(k\partial T/\partial\varphi) + \partial/\partial z(k\partial T/\partial z) + \dot{e} = \rho c\partial T/\partial t$. Laplace/Poisson/diffusion forms.

BCs: temperature, flux, insulated, convection, radiation, interface. Generation = conversion, not creation. Contact resistance $R_c = \Delta T/(Q/A)$: roughness, pressure, hardness, fluid in gaps.

Resistances (A2)

Wall L/kA; film $1/hA$; cylinder $\ln(r_2/r_1)/2\pi kL$; sphere $(r_2 - r_1)/4\pi k r_1 r_2$; $Q = \Delta T/\Sigma R$; $1/U = 1/h_1 + L/k + 1/h_2$.

Wire with $\dot{e}$: $T_0 - T_s = \dot{e}r_0^2/4k$, $Q = \dot{e}\pi r_0^2 L$. $k = k_0(1 + \beta t) \Rightarrow k_m$ at mean temperature.

Answers: wall 85 °C/6300 W; wall + film 7.39 kW; brick 518 W; steam pipe 121 W/m, ΔT_{ins} 284 °C; wire 149.4/152.7 °C.

Critical radius (A3)

$R = \ln(r_2/r_1)/2\pi kL + 1/2\pi r_2 hL$; $dR/dr_2 = 0 \Rightarrow$ **$r_{cr} = k/h$** (cylinder), **$2k/h$** (sphere). Q is max at r_{cr}. Below r_{cr} insulation increases loss (good for electric wires); removing sub-critical insulation reduces loss.

Fins (A4)

$d^2\theta/dx^2 - m^2\theta = 0$, $m = \sqrt{(hp/kA_c)}$ (pin $\sqrt{(4h/kD)}$).

Insulated tip: $\theta/\theta_b = \cosh m(L - x)/\cosh mL$, $Q = \sqrt{(hpkA_c)}\theta_b \tanh mL$. Long: tanh = 1. Corrected $L_c = L + A_c/p$.

$\eta = \tanh mL/mL$; $\varepsilon = Q_{fin}/hA_b\theta_b$ (> 2 to be worth it).

Longer: more effective, less efficient. Rod 25.1 W; parabolic pin η 0.985, 1.82 W; motor: bare 356 W ≥ 350 W (no fins needed: flag).

Transient (A5)

$Bi = hL_c/k$, $L_c = V/A$ (sphere D/6, cylinder D/4, plate L/2); $Bi < 0.1 \Rightarrow$ lumped. $\theta/\theta_i = e^{-bt}$, $b = hA/\rho Vc_p$, $bt = Bi\cdot Fo$.

One-term: $\theta_0 = A_1 e^{-\lambda_1^2\tau}$ ($Bi = hr_0/k$); $Q/Q_{max} = 1 - 2\theta_0 J_1(\lambda_1)/\lambda_1$. Semi-infinite: $(T - T_s)/(T_i - T_s) = \text{erf}(x/2\sqrt{\alpha t})$; contact T_s weighted by $\sqrt{(k\rho c_p)}$.

Copper ball Bi 0.00075; brass 448 °C; shaft T_0 390 °C, 15.9 MJ/m; pipe depth 0.65 m.

Boiling and condensation (A6)

Curve: natural (< 5 °C) → nucleate (5-30, CHF ≈ 1 MW/m² at C) → transition (30-120) → film (> 120, Leidenfrost at D). Burnout: q > CHF jumps to film boiling at ~1000 °C; design q << CHF.

Drop-wise h 5-10× film-wise, but design for film.

Horizontal tube $h_1 = 0.729[g\rho_l(\rho_l - \rho_v)h^*_{fg}k^3/\mu\Delta TD]^{1/4}$, N tubes $/N^{1/4}$, $h^* = h_{fg} + 0.68c_p\Delta T$; vertical 0.943 with L. $h_{hor}/h_{vert} = 0.77(L/D)^{1/4}$.

33 tubes: 3510 W/m²K, 0.0664 kg/s per m; 10×10: 4875, 3.68 MW, 1.50 kg/s; 15×15: 7150, 0.278 kg/s/m; film boiling 940 W/m.

Convection basics (B1)

Convection = conduction at the wall + fluid motion; $h = -k(\partial T/\partial y)_0/\Delta T \approx k/\delta_t$, so h falls as δ_t grows.

Velocity layer (v) vs thermal layer (α); $\delta/\delta_t \approx Pr^{1/3}$.

Turbulent: eddies, thin sublayer, steep gradient, high h.

$Re = VL/v$, $Pr = v/\alpha$, $Nu = hL/k$, $Gr = g\beta\Delta TL^3/v^2$, $Ra = GrPr$; $T_f = (T_s + T_\infty)/2$. Buckingham: 7 variables − 4 dimensions = 3 groups ⇒ $Nu = f(Re, Pr)$.

Boundary-layer analysis (B2)

Cubic profile ⇒ $\delta = 4.64x/\sqrt{Re_x}$ (Blasius 5.0); $C_{f,x} = 0.664/\sqrt{Re_x}$, $C_f = 1.328/\sqrt{Re_L}$. $\delta_t = 4.53x/Re^{1/2}Pr^{1/3} \Rightarrow Nu_x = 0.332Re_x^{1/2}Pr^{1/3}$, average 0.664.

Energy eq: $u\partial T/\partial x + v\partial T/\partial y = \alpha\partial^2 T/\partial y^2$. Integral: $d/dx\int u(T_\infty - T)dy = \alpha(\partial T/\partial y)_0$. Colburn: $StPr^{2/3} = C_f/2$. e^{-ay} profile: $h = ka = 96.8$. $h_x = ax^{-0.1}$: $\bar{h}/h = 1.11$.

Flat-plate numericals (B3)

Laminar ($Re < 5\times10^5$) $Nu = 0.664Re^{0.5}Pr^{1/3}$; turbulent $0.037Re^{0.8}Pr^{1/3}$. Low P: only ρ changes. Unheated start: $\times[1-(\xi/x)^{3/4}]^{-1/3}$.

Plastic sheet 437 + 548 = 985 W; 6 kPa air 569 W/m; heated after 12 cm: 262 W/m.

Internal flow (B4)

$Re < 2300$ laminar, $f = 64/Re$, $L_h = 0.05ReD$, $L_t = PrL_h$. Turbulent $Nu = 0.023Re^{0.8}Pr^n$ (0.4 heating, 0.3 cooling). $St = f/8$. Fully developed + uniform q: $\partial T/\partial x = dT_m/dx$ = const. Roughness affects only turbulent f.

$T_e = T_s - (T_s - T_i)e^{-hA/\dot{m}c_p}$; $Q = hA\Delta T_{lm}$; $\Delta P = f(L/D)\rho V^2/2$. Annulus h 8200; oil T_e 19.71 °C, 67.5 kW, 16.8 kW pump (inlet 20 °C assumed).

Tube banks (B5)

$V_{max} = S_TV/(S_T - D)$ (staggered: check $S_D < (S_T + D)/2$). In-line $Nu = 0.27Re^{0.63}Pr^{0.36}(Pr/Pr_s)^{0.25}$; staggered $0.35(S_T/S_L)^{0.2}Re^{0.6}$...; rows < 16 × F. $\Delta P = N_Lf\chi\rho V_{max}^2/2$.

Geothermal: h 92.2, T_e 29.1 °C, 24.9 kW/m, 21 Pa. In-line air: 69.4 °C, 87.7 kW, 0.41 kPa. Water staggered: 8190, 104 rows. Incropera: h 139, 19.8 kW/m.

Natural convection (B6)

$\beta = 1/T_f$; vertical $0.59Ra^{1/4}$ (L = height); horizontal $L_c = A/p$, hot up $0.54Ra^{1/4}$ ($0.15Ra^{1/3} > 10^7$), hot down $0.27Ra^{1/4}$; cylinder Churchill-Chu.

0.6 m plate: vertical 99 W (115 W Churchill-Chu), up 128 W (139 W), down 64 W. 8 cm pipe: 442 W.

Mass transfer (B7)

Fick: $J_A = -CD_{AB}dy_A/dx$; $D_{AB} \propto T^{3/2}/P$. Analogy: T↔C, k↔D, Pr↔Sc, Nu↔Sh. Homogeneous medium: no gradient, no diffusion. Absorption = bulk, adsorption = surface.

Equimolar: $N_A = CD(y_1 - y_2)/L$. Stagnant gas: $N_A = (CD/L)\ln[(1-y_L)/(1-y_0)]$. Sphere: $4\pi r_1r_2D\Delta C/(r_2 - r_1)$.

H_2 2.48×10^{-10} kg/s; toluene 5.49×10^{-7} kg/m²s; beaker 3.48×10^{-8} kg/s.

Abbreviations used in this pack

Every short form, written out. The answers also spell each one out the first time it appears.

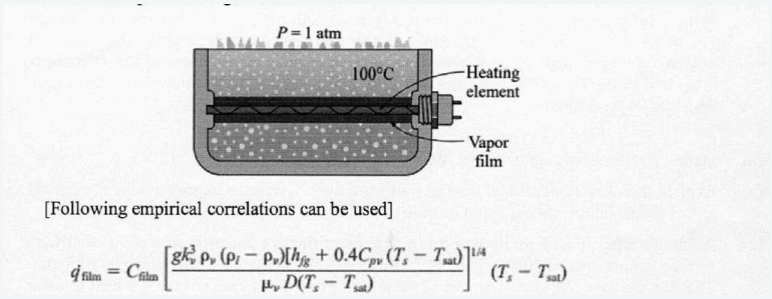
Short form	Full meaning
BC	boundary condition
Bi	Biot number
CHF	critical heat flux
Fo	Fourier number
Gr	Grashof number
ID	inner diameter
kmol	kilomole
kPa	kilopascal
kW	kilowatt
LMTD	log mean temperature difference
Nu	Nusselt number
OD	outer diameter
Pr	Prandtl number
Ra	Rayleigh number
Re	Reynolds number
Sc	Schmidt number
Sh	Sherwood number
St	Stanton number

Section A: Conduction, insulation, fins, transient, boiling and condensation

A6 Boiling and condensation Pass set Very likely · 100% Easy

PAST-PAPER QUESTIONS

- **2023 Q4(a)** [10] Differentiate between film and dropwise condensation; describe film condensation on a vertical plate.
- **2023 Q4(b)** [10] Describe pool boiling with the diagram, mentioning each stage.
- **2023 Q4(c)** [15] Same condenser (33 tubes per column, 1.5 cm, 12.35 kPa, 20 °C).
- **2022 Q4(b)** [7] For condensation over a vertical plate, which mode is preferable and why? Explain with equations.
- **2022 Q4(a)** [10] Explain briefly the physical mechanism of boiling with a diagram.
- **2022 Q4(c)** [18] Condenser columns of 33 tubes, D = 1.5 cm, 12.35 kPa, tube surfaces 20 °C. Find (i) average h, (ii) condensation rate per unit length of a column.
- **2021 Q4(a)** [8] Distinguish between film-wise and drop-wise condensation. Which is more effective?
- **2021 Q4(b)** [15] Draw the boiling curve, identify regimes and explain each.
- **2021 Q4(c)** [12] Horizontal polished copper element D = 5 mm, ε = 0.05, at 350 °C in water at 1 atm: heat transfer per unit length (film boiling).



[Following empirical correlations can be used]

$$\dot{q}_{film} = C_{film}\left[\frac{gk_v^3\rho_v(\rho_l-\rho_v)[h_{fg}+0.4C_{pv}(T_s-T_{sat})]}{\mu_v D(T_s-T_{sat})}\right]^{1/4}(T_s-T_{sat})$$

- **2020 Q2(b)** [5] Explain why condenser tubes are usually horizontal.
- **2020 Q3(b)** [15] Draw the boiling curve, identify the regimes and the burnout point. How is burnout caused and why is it avoided in boilers?
- **2020 Q3(a)** [15] Saturated steam at 85 °C condenses on 225 horizontal tubes (15 × 15), D = 1.27 cm, surface 75 °C; properties at 80 °C given. Condensation rate per m.
- **2019 Q4(a)** [8] Distinguish between (i) film-wise and drop-wise condensation, (ii) boiling and evaporation.
- **2019 Q4(b)** [7] Draw the pool boiling curve and show its regions.
- **2019 Q4(c)** [15] Condenser at 4.25 kPa, 100 horizontal tubes (10 × 10), 8 m long, D = 3 cm, surfaces at 20 °C. Find (i) heat transfer rate, (ii) condensation rate.

★ REMEMBER IT

- Boiling curve regions in order: **natural convection → nucleate (isolated bubbles, then columns) → peak heat flux (critical / burnout point) → transition → film boiling (Leidenfrost)**. Burnout: past the peak a vapour film blankets the surface, h collapses and the tube temperature jumps to melting.
- Dropwise condensation gives ~10× higher h than filmwise (no liquid film insulating the surface), but is hard to keep. Horizontal tubes condense better than vertical (thinner film,

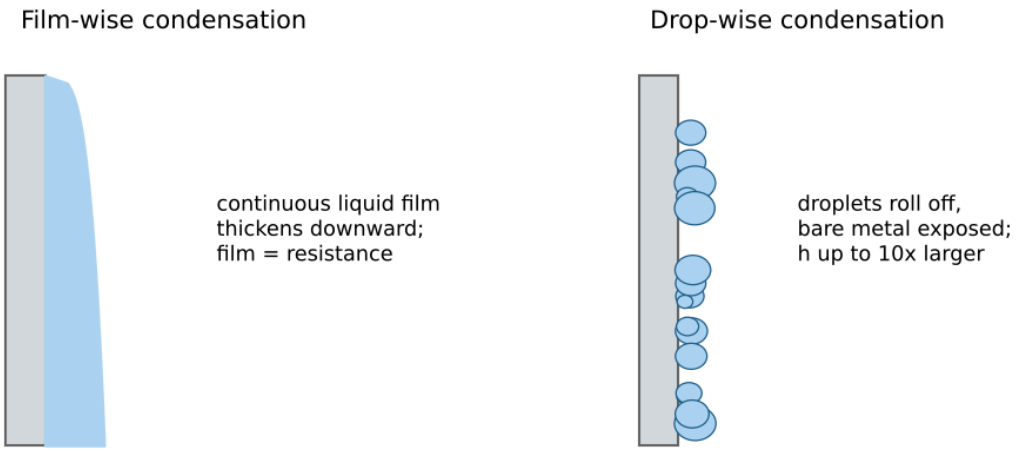
shorter path), $h_{horiz}/h_{vert} \approx 0.729/0.943\ (L/D)^{1/4}$.

Before you start: boiling and condensation in plain words

Boiling is evaporation at a solid-liquid interface when the surface is hotter than the saturation temperature of the liquid ($T_s > T_{sat}$): vapour bubbles form on the surface. **Condensation** is the reverse: vapour touching a surface colder than T_{sat} turns into liquid. Both involve latent heat, so h is very large (thousands of W/m^2K).

Excess temperature $\Delta T_e = T_s - T_{sat}$ for boiling, $T_{sat} - T_s$ for condensation. h_{fg} = latent heat (kJ/kg).

Film-wise vs drop-wise condensation (2019 4a, 2021 4a, 2023 4a, 2022 4b)



Film-wise and drop-wise condensation on a vertical surface

Film-wise	Drop-wise
Condensate wets the surface and forms a continuous liquid film	Condensate forms droplets that do not wet the surface and roll off
The film is a thermal resistance: vapour must conduct through it	Much of the surface is bare: vapour touches the metal directly
h about 5-20 kW/m^2K	h 5-10 times larger (up to 100+ kW/m^2K)
Occurs on clean, wettable surfaces; stable, predictable; used for design	Needs non-wetting coatings (Teflon, gold, oils) that wear off; hard to sustain

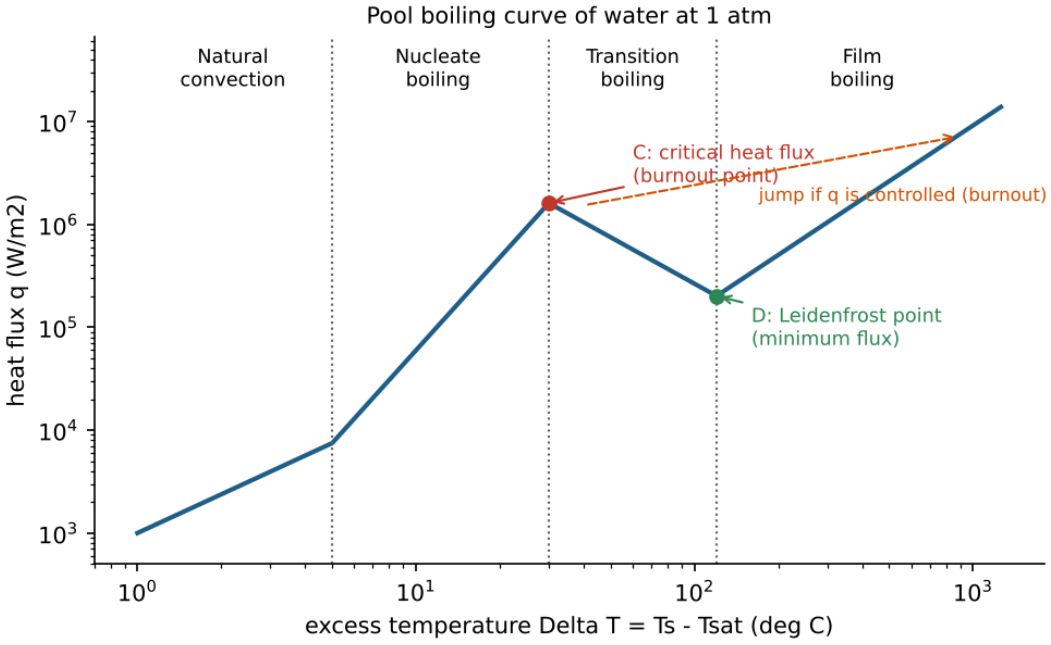
Which is more effective / preferable? Drop-wise condensation gives much higher heat transfer coefficients, so it is more effective. But it cannot be maintained for long in practice, so condensers are **designed for film condensation** (conservative). On a vertical plate the Nusselt film theory gives $h = 0.943[g\rho_l(\rho_l - \rho_v)h^*_{fg}k_l^3/\mu_l(T_{sat} - T_s)L]^{1/4}$: the film thickens downward ($\delta \propto x^{1/4}$) and h falls ($h \propto 1/\delta$). Drop-wise condensation avoids this film resistance entirely.

Film condensation on a vertical plate (2023 4a): vapour condenses at the top, the liquid film flows down under gravity, growing thicker as more vapour condenses on it; flow is laminar near the top and may become wavy/turbulent further down (judged by the film Reynolds number). Heat passes by conduction through the film from the liquid-vapour interface (at T_{sat}) to the wall (T_s).

Boiling vs evaporation (2019 4a): evaporation occurs at a liquid-vapour (free) interface when the vapour pressure is less than the saturation pressure of the liquid at its temperature, with no bubbles, at any temperature (e.g. a wet road drying). Boiling occurs at a solid-liquid interface when the surface is above T_{sat}, with bubble formation and very high heat flux.

Why condenser tubes are horizontal (2020 2b): for a single tube $h_{horizontal}/h_{vertical} = 0.729/0.943 \times (L/D)^{1/4} = 0.77(L/D)^{1/4}$. For L/D = 100, this is 2.4: the condensate film on a horizontal tube is thin because it drains over a short path (around the circumference, length ~D) instead of along the whole tube length L. Horizontal tubes also make the condenser shell compact and easy to drain.

Pool boiling curve (2019 4b, 2020 3b, 2021 4b, 2023 4b, 2022 4a)



Boiling curve for water at 1 atm (Nukiyama)

1. **Natural convection boiling (ΔT_e < 5 °C):** liquid near the surface is slightly superheated; it rises by natural convection and evaporates at the free surface; no bubbles on the heater.
2. **Nucleate boiling (5-30 °C, A-C):** bubbles form at nucleation sites (cavities, scratches). A-B: isolated bubbles that collapse in the liquid; B-C: bubbles form jets and columns that rise to the surface. Bubble motion stirs the liquid strongly, so q rises steeply. This is the most desirable regime: high q at small ΔT. Point C is the maximum: **critical heat flux (CHF)**, about 1 MW/m^2 for water.
3. **Transition boiling (30-120 °C, C-D):** part of the surface is covered by a vapour film (a poor conductor) which forms and collapses unsteadily; q *decreases* as ΔT increases.
4. **Film boiling (> 120 °C, beyond D):** a stable vapour blanket covers the surface; heat passes by conduction and radiation through the vapour. Point D (minimum heat flux) is the **Leidenfrost point**. q rises again at very high ΔT as radiation grows.

Burnout: in electric heaters and nuclear reactors the heat flux q (not the temperature) is controlled. If q is raised beyond the CHF (point C), the surface cannot transfer it by nucleate boiling, a vapour film forms, and the operating point jumps horizontally to the film-boiling curve at the same q, where ΔT_e is about 1000 °C. The surface temperature exceeds the melting point of the material and it burns out. **In boiler design** the heat flux is therefore kept well below CHF (with a safety margin), so

that tubes stay in the nucleate regime: high heat transfer, low wall temperature, and no risk of tube rupture.

Physical mechanism of boiling (2022 4a): a vapour bubble can form only on a surface hotter than T_{sat} and only at nucleation sites where small gas or vapour pockets exist. A bubble grows by evaporation at its base (microlayer), departs under buoyancy when its size reaches a critical diameter, and the cooler liquid rushes in to fill its place: this cycle of bubble formation, growth, departure and liquid replacement removes heat from the surface at very high rates.

Numerical 1: condenser, 33 tubes per column (2022 4c, 2023 4c)

Step (write this)	Why / what it means
12.35 kPa ⇒ T_{sat} = 50 °C; h_{fg} = 2383 kJ/kg; ρ_v = 0.0831 kg/m^3	Saturation table.
Liquid at film temperature T_f = (50 + 20)/2 = 35 °C: ρ_l = 994 kg/m^3, μ_l = 0.720×10^{-3} kg/m s, k_l = 0.623 W/m K, c_{pl} = 4178 J/kg K	Water table (Cengel A-9).
$h^*_{fg} = h_{fg} + 0.68c_{pl}(T_{sat} - T_s)$ = 2383 + 0.68 × 4.178 × 30 = 2468 kJ/kg	Adds the subcooling of the liquid film.
$h_{1\ tube}$ = 0.729[9.81 × 994(994 - 0.08) × 2.468×10^6 × 0.623^3/(0.720×10^{-3} × 30 × 0.015)]$^{1/4}$ = 8415 W/m^{2}K	Single horizontal tube.
(i) $h_{33\ tubes} = h_1/N^{1/4} = 8415/33^{0.25}$ = 3510 W/m^{2}K	Lower tubes are covered by condensate dripping from above.
Q per metre of column = $hN\pi D(T_{sat} - T_s)$ = 3510 × 33 × π × 0.015 × 30 = 163.8 kW/m	
(ii) $\dot{m} = Q/h^*_{fg}$ = 163.8/2468 = 0.0664 kg/s per m (≈ 239 kg/h)	

Numerical 2: power-plant condenser, 10 × 10 tubes (2019 4c)

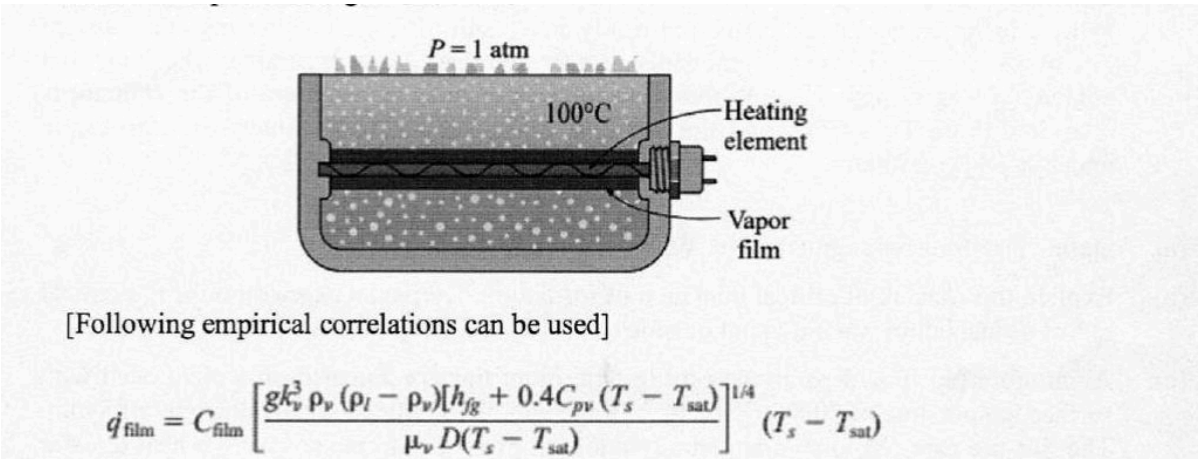
Step (write this)	Why / what it means
4.25 kPa ⇒ T_{sat} = 30 °C, h_{fg} = 2431 kJ/kg, ρ_v = 0.030 kg/m^3	Saturation table.
T_f = 25 °C: ρ_l = 997, μ_l = 0.891×10^{-3}, k_l = 0.607, c_{pl} = 4180	Liquid properties.
h^*_{fg} = 2431 + 0.68 × 4.18 × 10 = 2459 kJ/kg	
h_1 = 0.729[9.81 × 997 × 997 × 2.459×10^6 × 0.607^3/(0.891×10^{-3} × 10 × 0.03)]$^{1/4}$ = 8670 W/m^{2}K	
$h_{10\ tubes}$ = 8670/10$^{0.25}$ = 4875 W/m^{2}K	N = 10 tubes in each vertical column.
A = 100πDL = 100 × π × 0.03 × 8 = 75.4 m^2	
(i) Q = hAΔT = 4875 × 75.4 × 10 = 3.68 MW	
(ii) $\dot{m}$ = 3.68×10^6/2.459×10^6 = 1.50 kg/s	

Numerical 3: 15 × 15 tube bundle with given properties (2020 3a)

Step (write this)	Why / what it means

$\Delta T = 85 - 75 = 10\ °C$; $N = 15$ tubes per column; $\rho_v(85\ °C) = 0.354\ kg/m^3$	Only liquid properties are given; c_p not given, so h_{fg} is used as given.
$h = 0.725[g\rho_l(\rho_l - \rho_v)h_{fg}k^3/(\mu N\ D\Delta T)]^{1/4} = 0.725[9.81 \times 974 \times 973.6 \times 2.309\times10^6 \times 0.668^3/(0.355\times10^{-3} \times 15 \times 0.0127 \times 10)]^{1/4}$	Nusselt correlation for N horizontal tubes in a column (Holman uses 0.725).
$h = 7150\ W/m^2K$	
Q per $m = h(225\pi D)\Delta T = 7150 \times 225 \times \pi \times 0.0127 \times 10 = 642\ kW/m$	
$\dot{m} = 642/2309 = 0.278$ kg/s per metre of bundle (≈ 1000 kg/h)	

Numerical 4: film boiling on a heating wire (2021 4c)



Question figure and correlation (2021 paper)

Step (write this)	Why / what it means
$\Delta T_e = 350 - 100 = 250\ °C > 120\ °C \Rightarrow$ film boiling	Check the regime on the boiling curve.
Liquid at 100 °C: $\rho_l = 957.9\ kg/m^3$, $h_{fg} = 2257\ kJ/kg$. Vapour at $T_f = 225\ °C$, 1 atm: $\rho_v = 0.443$, $c_{pv} = 1981\ J/kg\ K$, $k_v = 0.0359\ W/m\ K$, $\mu_v = 1.72\times10^{-5}\ kg/m\ s$	Vapour properties at the film temperature.
$q_{film} = 0.62[9.81 \times 0.0359^3 \times 0.443(957.9 - 0.443)(2.257\times10^6 + 0.4 \times 1981 \times 250)/(1.72\times10^{-5} \times 0.005 \times 250)]^{1/4} \times 250$	$C_{film} = 0.62$ for horizontal cylinders.
$q_{film} = 5.96\times10^4\ W/m^2$	
$q_{rad} = \varepsilon\sigma(T_s^4 - T_{sat}^4) = 0.05 \times 5.67\times10^{-8}(623^4 - 373^4) = 373\ W/m^2$	Polished copper: low emissivity, so little radiation.
$q_{total} = q_{film} + \frac{3}{4}q_{rad} = 59{,}600 + 280 = 59{,}880\ W/m^2$	Radiation and film conduction interact; Cengel combines them this way.
Q per metre $= q\pi D = 59{,}880 \times \pi \times 0.005 = 940$ W/m	Cengel Example 10-3.

Memory trick: Boiling curve: "**N-N-T-F**" (Natural, Nucleate, Transition, Film) with **C**ritical flux at the top and **L**eidenfrost at the bottom. Condenser: $h_N = h_1/N^{1/4}$, $h^* = h_{fg} + 0.68c_p\Delta T$.

A2 Steady one-dimensional conduction: walls, cylinders, spheres, thermal resistance, heat generation Pass set Very likely · 100% Easy

PAST-PAPER QUESTIONS

- **2023 Q2(c)** [11] Same brick wall (3 m × 6 m × 0.25 m, k = 0.8, 14 °C and 5 °C): rate of heat loss.
- **2023 Q1(c)** [13] Plane wall with $k = k_0t + 0.5k_0t^2$ and faces at t_1, t_2: (i) derive the steady heat flow, (ii) temperature at which the mean k should be evaluated.
- **2023 Q2(a)** [12] What is the overall heat transfer coefficient? Deduce its expression.
- **2022 Q1(c)** [13] Plane wall L = 0.4 m, k = 1.8 W/m K, A = 30 m^2; left face 90 °C, right face loses heat by convection to 25 °C air with h = 24 W/m^2K. Find the heat transfer rate.
- **2022 Q2(d)** [7] Brick wall 3 m × 6 m × 0.25 m, k = 0.8 W/m K, surfaces at 14 °C and 5 °C. Find the heat loss.
- **2021 Q1(c)** [17] Resistance wire r = 0.2 cm, k = 15 W/m K, $\dot{e}$ = 50 W/cm^3, inside 0.5 cm ceramic (k = 1.2); outer surface 45 °C. Find centreline and interface temperatures.
- **2020 Q1(a)** [15] Hollow sphere with surfaces at T_1 (r_1) and T_2 (r_2), constant k: develop the radial heat flow rate.
- **2019 Q1(c)** [14] Plane wall L = 0.2 m, k = 1.2 W/m K, A = 15 m^2, T_1 = 120 °C, T_2 = 50 °C. Find (i) T at x = 0.1 m, (ii) heat conduction rate.
- **2019 Q2(c)** [15] Steam at 320 °C in a cast iron pipe (k = 80), D_1 = 5 cm, D_2 = 5.5 cm, 3 cm glass wool (k = 0.05); h_i = 60, h_o = 18 W/m^2K, surroundings 5 °C. Find heat loss per m and temperature drops across pipe and insulation.

LEARN IT FROM ZERO

Thermal resistance, from zero

Heat flowing through a wall behaves exactly like electric current through a resistor: **Q = ΔT / R**. Temperature difference is the "voltage", heat rate the "current". Each layer and each surface film has its own resistance, and layers one after another simply add (series).

- Plane wall: R = L/(kA)
- Convection film: R = 1/(hA)
- Cylinder layer: $R = \ln(r_2/r_1)/(2\pi kL)$
- Sphere shell: $R = (r_2 - r_1)/(4\pi k r_1 r_2)$

Once you have Q, any inside temperature comes from Q = ΔT/R across the part you want.

CALCULATOR **Plane wall with convection on one side (2022 1(c) type)**

Change any number; every step is recalculated and shown with the numbers put in.

1. Wall resistance: R = L/kA = **0.4** / (**1.8** × **30**) = **0.007407** K/W
2. Convection resistance: R = 1/hA = 1 / (**24** × **30**) = **0.001389** K/W
3. Total: **0.007407** + **0.001389** = **0.008796** K/W
4. Heat rate: Q = ΔT/R = (**90** - **25**) / **0.008796** = **7389.5** W

5. Right face temperature: $T_2 = T_1 - QR_{wall}$ = **90** - **7389** × **0.007407** = **35.26** °C

CALCULATOR **Insulated pipe: composite cylinder per metre length (2019 2(c) type)**

Change any number; every step is recalculated and shown with the numbers put in.

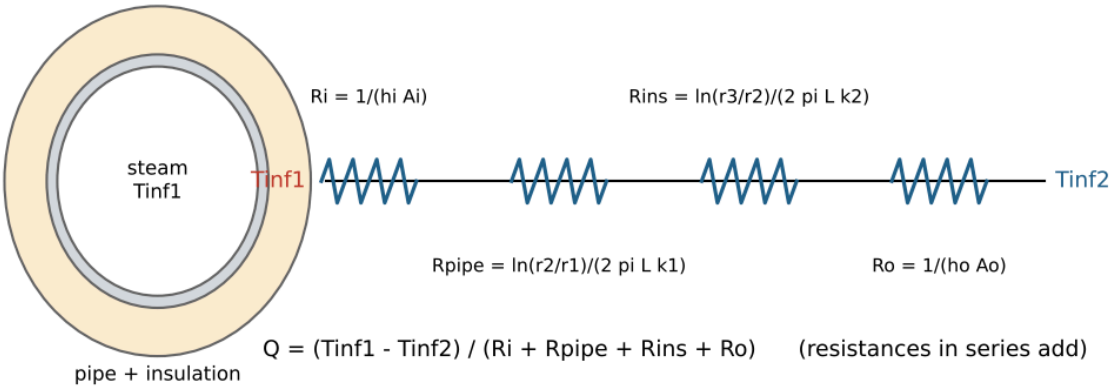
1. Inside film: 1 / (**60** × 2 × π × **0.025**) = **0.10610** K/W
2. Pipe wall: ln(**0.0275** / **0.025**) / (2 × π × **80**) = **1.896 × 10^{-4}** K/W
3. Insulation: ln(**0.0575** / **0.0275**) / (2 × π × **0.05**) = **2.3479** K/W
4. Outside film: 1 / (**18** × 2 × π × **0.0575**) = **0.1538** K/W
5. Heat loss per metre: (**320** - **5**) / (**0.1061** + **1.896 × 10^{-4}** + **2.348** + **0.1538**) = **120.8** W/m
6. Outer insulation surface: **5** + **120.8** × **0.1538** = **23.6** °C

Defaults are the 2019 2(c) numbers: D_1 = 5 cm, D_2 = 5.5 cm, 3 cm glass wool.

Before you start: thermal resistance: the electrical analogy

For steady one-dimensional conduction, Q = ΔT/R exactly like current I = ΔV/R. Each layer or surface film has a resistance (K/W) and resistances in series add. This turns most wall and pipe problems into one line of arithmetic.

Plane wall R = L/(kA); convection film R = 1/(hA); cylinder layer R = $\ln(r_2/r_1)/(2\pi kL)$; spherical shell R = $(r_2 - r_1)/(4\pi k\, r_1 r_2)$.



Resistance network for a steam pipe with insulation

1. Plane wall: temperature and heat rate (2019 1c)

Step (write this)	Why / what it means
Steady, no generation, constant k: $d^2T/dx^2 = 0 \Rightarrow T(x) = C_1x + C_2$	Conduction equation reduces to this; integrate twice.
BCs $T(0) = T_1$, $T(L) = T_2 \Rightarrow$ **$T(x) = T_1 - (T_1 - T_2)x/L$**	Linear profile.
(i) $T(0.1) = 120 - 70 \times 0.1/0.2$ = **85 °C**	Mid-plane is the average because the profile is linear.

(ii) $Q = kA(T_1 - T_2)/L = 1.2 \times 15 \times 70/0.2 =$ **6300 W**	Fourier's law.

2. Wall with convection on one side (2022 1c)

Step (write this)	Why / what it means
$R_{wall} = L/(kA) = 0.4/(1.8 \times 30) = 0.007407$ K/W	Conduction resistance.
$R_{conv} = 1/(hA) = 1/(24 \times 30) = 0.001389$ K/W	Convection film on the right.
$R_{total} = 0.008796$ K/W	Series.
$Q = (90 - 25)/0.008796 = 7390$ W ≈ 7.39 kW	Right-face temperature = 90 − 7390 × 0.007407 = 35.3 °C.

3. Brick wall (2022 2d, 2023 2c)

Step (write this)	Why / what it means
$A = 3 \times 6 = 18$ m^2	
$Q = kA\Delta T/L = 0.8 \times 18 \times (14 - 5)/0.25 = 518.4$ W	Steady one-dimensional conduction; both surface temperatures known.

4. Insulated steam pipe (2019 2c)

Step (write this)	Why / what it means
$r_1 = 2.5$ cm, $r_2 = 2.75$ cm, $r_3 = 2.75 + 3 = 5.75$ cm; $L = 1$ m	Radii from diameters and insulation thickness.
$R_i = 1/(h_i 2\pi r_1 L) = 1/(60 \times 2\pi \times 0.025) = 0.1061$ K/W	Inner convection.
$R_{pipe} = \ln(r_2/r_1)/(2\pi k_1 L) = \ln(1.1)/(2\pi \times 80) = 0.00019$ K/W	Cast iron: tiny resistance.
$R_{ins} = \ln(r_3/r_2)/(2\pi k_2 L) = \ln(2.091)/(2\pi \times 0.05) = 2.348$ K/W	Glass wool dominates.
$R_o = 1/(h_o 2\pi r_3 L) = 1/(18 \times 2\pi \times 0.0575) = 0.1538$ K/W	Outer combined convection + radiation.
$R_{total} = 2.608$ K/W; **(i) $Q = (320 - 5)/2.608 =$ 121 W per m**	
(ii) $\Delta T_{pipe} = QR_{pipe} = 0.02$ °C; $\Delta T_{ins} = QR_{ins} =$ 284 °C	Almost all the temperature drop is across the insulation.

5. Wire with heat generation in a ceramic sleeve (2021 1c)

Before you start: heat generation in a long cylinder

Steady, uniform generation $\dot{e}$ in a wire of radius r_0: $(1/r)d/dr(r\,dT/dr) + \dot{e}/k = 0$ gives $T(r) = T_s + \dot{e}(r_0^2 - r^2)/(4k)$, so $T_{centre} - T_{surface} = \dot{e}r_0^2/(4k)$. All the heat generated must leave through the surface: $Q = \dot{e}\pi r_0^2 L$.

Step (write this)	Why / what it means
$\dot{e} = 50$ W/cm^3 $= 50 \times 10^6$ W/m^3; $r_1 = 0.002$ m, $r_2 = 0.002 + 0.005 = 0.007$ m	1 cm^3 = 10^{-6} m^3.

Q per metre = $\dot{e}\pi r_1^2 = 5\times10^7 \times \pi \times (0.002)^2 = 628.3$ W/m	Energy balance on the wire.
Interface: $T_1 = T_s + Q \ln(r_2/r_1)/(2\pi k_c) = 45 + 628.3 \times \ln 3.5/(2\pi \times 1.2)$	The ceramic has no generation: plain cylindrical resistance.
$T_{interface} = 45 + 104.4 = 149.4$ °C	
$T_{centre} = T_1 + \dot{e}r_1^2/(4k_w) = 149.4 + 5\times10^7 \times 4\times10^{-6}/60 = 152.7$ °C	Wire rise is only 3.3 °C because the wire conducts well.

6. Hollow sphere: radial heat flow (2020 1a)

Step (write this)	Why / what it means
Steady, no generation, constant k: $(1/r^2)\, d/dr(r^2 dT/dr) = 0$	Spherical form of the conduction equation.
$r^2 dT/dr = C_1 \Rightarrow T(r) = -C_1/r + C_2$	Integrate twice.
$T(r_1) = T_1$, $T(r_2) = T_2 \Rightarrow C_1 = (T_1 - T_2)r_1r_2/(r_1 - r_2)$	Apply BCs and subtract.
$Q = -k(4\pi r^2)dT/dr = -4\pi k C_1$	Fourier with sphere area; the same at every r.
$Q = 4\pi k\, r_1r_2(T_1 - T_2)/(r_2 - r_1) \Rightarrow R_{sphere} = (r_2 - r_1)/(4\pi k\, r_1r_2)$	

7. Variable conductivity $k = k_0t + 0.5k_0t^2$ (2023 1c)

Note: the conductivity is printed as $k = k_0t + 0.5k_0t^2$; it is solved as printed. (If the intended law is the usual linear one, $k = k_0(1 + \beta t)$, the same method gives the mean temperature $(t_1 + t_2)/2$.)

Step (write this)	Why / what it means
$Q = -kA\, dt/dx \Rightarrow Q\, dx = -A\, k(t)\, dt$	Steady: Q is the same at every x, so separate variables.
$\int_0^L Q\, dx = -A\, k_0 \int_{t1}^{t2} (t + 0.5t^2) dt$	
(i) $Q = (Ak_0/L)[(t_1^2 - t_2^2)/2 + (t_1^3 - t_2^3)/6]$	Steady heat flow through the wall.
Simplified Fourier form: $Q = k_mA(t_1 - t_2)/L \Rightarrow k_m = k_0[(t_1 + t_2)/2 + (t_1^2 + t_1t_2 + t_2^2)/6]$	Divide (i) by $(t_1 - t_2)$ using $a^2 - b^2$ and $a^3 - b^3$ factorisations.
(ii) Mean temperature t_m satisfies $k(t_m) = k_m$: $t_m + 0.5t_m^2 = (t_1 + t_2)/2 + (t_1^2 + t_1t_2 + t_2^2)/6$	
$t_m = -1 + \sqrt{[1 + (t_1 + t_2) + (t_1^2 + t_1t_2 + t_2^2)/3]}$	Positive root of the quadratic $0.5t_m^2 + t_m - \text{RHS} = 0$.

8. Overall heat transfer coefficient (2023 2a)

Definition: U (W/m^{2}K) is the heat transfer rate per unit area per unit overall temperature difference between two fluids separated by a wall: $Q = UA(T_{\text{hot fluid}} - T_{\text{cold fluid}})$. It combines all the resistances in one number.

Step (write this)	Why / what it means

Hot fluid → wall: $Q = h_1A(T_{\infty 1} - T_1) \Rightarrow T_{\infty 1} - T_1 = Q/(h_1A)$	Convection on the hot side.
Through wall: $Q = kA(T_1 - T_2)/L \Rightarrow T_1 - T_2 = QL/(kA)$	Conduction.
Wall → cold fluid: $T_2 - T_{\infty 2} = Q/(h_2A)$	Convection on the cold side.
Add: $T_{\infty 1} - T_{\infty 2} = (Q/A)(1/h_1 + L/k + 1/h_2)$	Intermediate temperatures cancel.
$Q = UA(T_{\infty 1} - T_{\infty 2})$, $1/U = 1/h_1 + L/k + 1/h_2$	For several layers add $\Sigma L_i/k_i$; for a tube, $UA = 1/R_{total}$.

How it is measured: in a test (e.g. a heat exchanger or a wall sample between two fluids), measure the heat rate Q (from $\dot{m}c_p\Delta T$ of one fluid or the electrical heater power), the area A and the two fluid temperatures; then $U = Q/(A\Delta T)$ (for an exchanger use ΔT_{lm}).

Memory trick: "Q = ΔT / ΣR": wall L/kA, film 1/hA, pipe $\ln(r_o/r_i)/2\pi kL$, sphere $(r_o - r_i)/4\pi kr_ir_o$. Wire: centre is hotter by $\dot{e}r^2/4k$.

WRITE IN EXAM **Write this in the exam (this much is enough for full marks)**

2022 1(c) Given: L = 0.4 m, k = 1.8 W/m K, A = 30 m², left face 90 °C, right face to air at 25 °C with h = 24 W/m²K

Wall resistance: $R_{wall} = L/(kA)$

$= 0.4/(1.8 \times 30)$

$=$ **0.007407 K/W**

Convection resistance: $R_{conv} = 1/(hA)$

$= 1/(24 \times 30)$

$=$ **0.001389 K/W**

Heat rate: $Q = (T_1 - T_\infty)/(R_{wall} + R_{conv})$

$= 65/0.008796$

$=$ **7390 W**

Right-face temperature: $T_2 = T_1 - QR_{wall}$

$= 90 - 7390 \times 0.007407$

$=$ **35.3 °C**

A1 Conduction basics: modes, k and α, boundary conditions, general conduction equation

Pass set | Very likely · 87% | Easy

PAST-PAPER QUESTIONS

- **2023 Q1(a)** [8] What are the basic modes of heat transfer? Describe the modes by mentioning their governing equations.
- **2023 Q1(b)** [14] Derive the general heat conduction equation for a rectangular coordinate system.
- **2022 Q1(a)** [8] What are the basic modes of heat transfer? Write down the basic differences among them.
- **2022 Q1(b)** [14] Derive the general heat conduction equation in cylindrical coordinates.
- **2022 Q2(a)** [5] What is thermal contact resistance? Name the factors that affect it. (Critical thickness part in A3.)
- **2021 Q1(a)** [10] Explain various kinds of boundary conditions commonly considered in heat conduction problems.
- **2021 Q1(b)** [8] Does heat generation in a solid violate the first law of thermodynamics? Explain.
- **2019 Q1(a)** [6] What are thermal conductivity and thermal diffusivity?
- **2019 Q2(a)** [5] Distinguish between steady and transient heat transfer.
- **2019 Q1(b)** [15] Derive the general heat conduction equation for a cylindrical coordinate system.
- **2019 Q3(a)** [3] Define and explain thermal contact resistance (the critical-thickness part is in A3).

★ REMEMBER IT

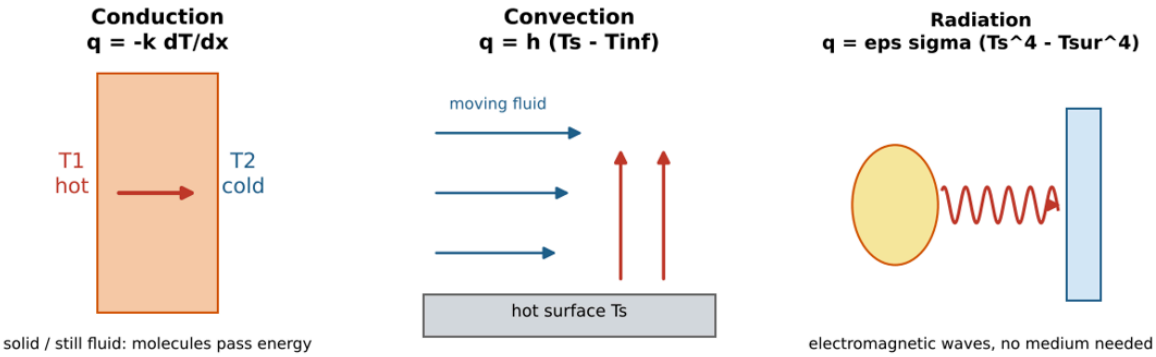
- Three modes: **conduction** $q = -kA\,dT/dx$ (touching, solids), **convection** $q = hA(T_s - T_\infty)$ (moving fluid), **radiation** $q = \varepsilon\sigma A(T_s^4 - T_{sur}^4)$ (no medium needed).
- k = how well a material conducts; $\alpha = k/\rho c_p$ = how fast heat *spreads* through it (k divided by how much heat it soaks up).
- Boundary conditions: specified temperature, specified heat flux (insulated = zero flux), convection, radiation, interface.

Before you start: what heat transfer is

Thermodynamics tells us *how much* energy moves from one state to another. **Heat transfer** tells us *how fast* it moves (watts) and what the temperature is at every point. Heat always flows from hot to cold, and the driving force is a **temperature difference**.

Units: heat rate Q in W; heat flux $q = Q/A$ in W/m^2; thermal conductivity k in W/m K; heat transfer coefficient h in W/m^2 K.

Modes of heat transfer and their governing equations (2022 1a, 2023 1a)



The three modes of heat transfer

	Conduction	Convection	Radiation
Mechanism	energy passed from more energetic to less energetic molecules by collisions/lattice vibration and free electrons	conduction combined with bulk motion of a fluid next to a surface	electromagnetic waves (photons) emitted by matter because of its temperature
Medium	solids, and liquids/gases at rest	moving liquid or gas	none needed (works in vacuum)
Law	Fourier: $Q = -kA\,dT/dx$	Newton: $Q = hA_s(T_s - T_\infty)$	Stefan-Boltzmann: $Q = \varepsilon\sigma A_s(T_s^4 - T_{surr}^4)$
Key property	k (W/m K)	h (W/m^2 K), depends on flow	emissivity ε (0-1), σ = 5.67×10^{-8} W/m^{2}K^4
Example	heat through a wall, a spoon in tea	fan cooling a CPU, wind on a wall	sunlight, heat from a fire

Side note: The minus sign in Fourier's law is there because heat flows down the temperature gradient: if T falls with x ($dT/dx < 0$), Q is positive in +x.

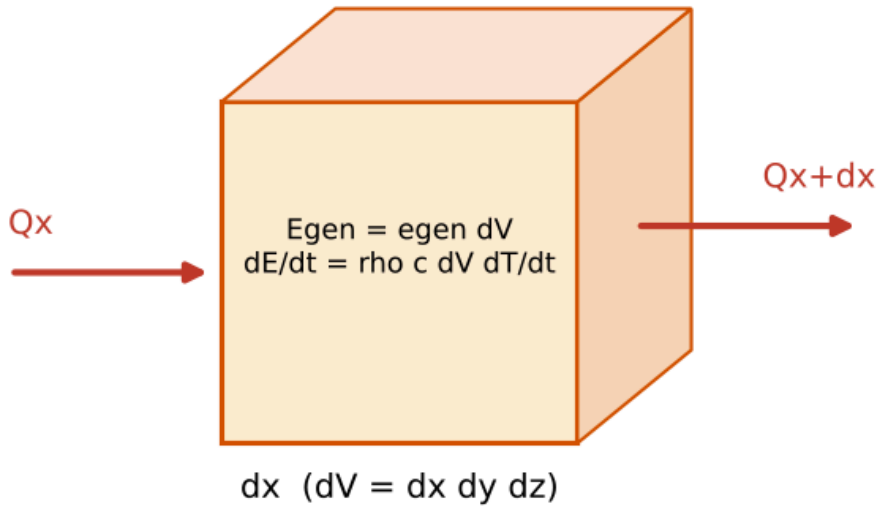
Thermal conductivity and thermal diffusivity (2019 1a)

- **Thermal conductivity k**: the rate of heat transfer through a unit thickness of material per unit area per unit temperature difference (W/m K). It measures how well a material *conducts* heat. Copper 401, steel 15-60, brick 0.7, glass wool 0.04, air 0.026 W/m K.
- **Thermal diffusivity $\alpha = k/(\rho c_p)$** (m^2/s): heat conducted divided by heat stored per unit volume. It measures how *fast* heat spreads through a material during transient heating. Large α (metals) ⇒ the body responds quickly to a temperature change; small α (wood, soil) ⇒ slow response.

Steady vs transient heat transfer (2019 2a)

Steady	Transient (unsteady)
Temperature at any point does not change with time: $\partial T/\partial t = 0$	Temperature changes with time: $\partial T/\partial t \neq 0$
Heat in = heat out; no energy stored	Part of the heat is stored (or released): energy content changes
Example: heat loss through a house wall on a day with constant conditions	Example: a hot shaft cooling, an egg boiling, a plate in an oven

General heat conduction equation: Cartesian coordinates (2023 1b)



Energy balance on a small rectangular element

Step (write this)	Why / what it means
Element dx × dy × dz with heat generation $\dot{e}_{gen}$ (W/m^3), density ρ, specific heat c	Choose a tiny volume and apply energy conservation to it.
$(Q_x + Q_y + Q_z) - (Q_{x+dx} + Q_{y+dy} + Q_{z+dz}) + \dot{E}_{gen} = dE_{element}/dt$	Rate in − rate out + generated = rate of change of stored energy.
$\dot{E}_{gen} = \dot{e}_{gen}$ dx dy dz; $dE/dt = \rho c$ dx dy dz $\partial T/\partial t$	Generation is per unit volume; storage is mass × c × rate of temperature rise.
Divide by dx dy dz: $-(Q_{x+dx} - Q_x)/(dy\ dz\ dx)$ - ... + $\dot{e}_{gen} = \rho c\ \partial T/\partial t$	
dx → 0: $(Q_{x+dx} - Q_x)/dx \to \partial Q_x/\partial x$, with $Q_x = -k$ dy dz $\partial T/\partial x$	Definition of a derivative + Fourier's law.
$\partial/\partial x(k\partial T/\partial x) + \partial/\partial y(k\partial T/\partial y) + \partial/\partial z(k\partial T/\partial z) + \dot{e}_{gen} = \rho c\ \partial T/\partial t$	General (three-dimensional) conduction equation.
Constant k: **$\partial^2T/\partial x^2 + \partial^2T/\partial y^2 + \partial^2T/\partial z^2 + \dot{e}_{gen}/k = (1/\alpha)\partial T/\partial t$**	Divide by k; α = k/ρc.

Special forms: steady (∂/∂t = 0) ⇒ **Poisson** equation $\nabla^2T + \dot{e}/k = 0$; transient with no generation ⇒ **diffusion (Fourier)** equation $\nabla^2T = (1/\alpha)\partial T/\partial t$; steady with no generation ⇒ **Laplace** equation $\nabla^2T = 0$.

General heat conduction equation: cylindrical coordinates (2019 1b, 2022 1b)

Step (write this)	Why / what it means
Element: dr × r dφ × dz; volume dV = r dφ dr dz	Cylindrical coordinates (r, φ, z) suit pipes and wires.
Areas normal to each direction: A_r = r dφ dz, A_φ = dr dz, A_z = r dφ dr	The area facing r grows with r: this is what makes the cylindrical equation different.
$Q_r = -k\ A_r\ \partial T/\partial r$, $Q_\varphi = -k\ A_\varphi\ (1/r)\partial T/\partial \varphi$, $Q_z = -k\ A_z\ \partial T/\partial z$	Fourier's law; distance along φ is r dφ.

Energy balance: $-(\partial Q_r/\partial r)dr - (\partial Q_\varphi/\partial \varphi)d\varphi - (\partial Q_z/\partial z)dz + \dot{e}_{gen}dV = \rho c\, dV\, \partial T/\partial t$	Same balance as the Cartesian case.
Substitute and divide by $dV = r\, d\varphi\, dr\, dz$:	
$(1/r)\partial/\partial r(kr\partial T/\partial r) + (1/r^2)\partial/\partial \varphi(k\partial T/\partial \varphi) + \partial/\partial z(k\partial T/\partial z) + \dot{e}_{gen} = \rho c\, \partial T/\partial t$	General conduction equation in cylindrical coordinates.
Steady, one-dimensional, no generation: $(1/r)\, d/dr(r\, dT/dr) = 0$	The form used for pipes: gives $T = C_1 \ln r + C_2$.

Spherical (for reference): $(1/r^2)\partial/\partial r(kr^2\partial T/\partial r) + (1/r^2\sin^2\theta)\partial/\partial\varphi(k\partial T/\partial\varphi) + (1/r^2\sin\theta)\partial/\partial\theta(k\sin\theta\,\partial T/\partial\theta) + \dot{e} = \rho c\,\partial T/\partial t$.

Boundary conditions (2021 1a)

Boundary condition (BC)	Mathematical form at x = 0	Example
1. Specified temperature	$T(0, t) = T_1$	surface in contact with melting ice or boiling water
2. Specified heat flux	$-k\,\partial T(0,t)/\partial x = \dot{q}_0$	electric heater on a surface
3. Insulated (adiabatic) / symmetry	$\partial T(0,t)/\partial x = 0$	well-insulated face; centre plane of a symmetric wall
4. Convection	$-k\,\partial T(0,t)/\partial x = h[T_\infty - T(0,t)]$	wall exposed to air
5. Radiation	$-k\,\partial T(0,t)/\partial x = \varepsilon\sigma[T_{surr}^4 - T(0,t)^4]$	spacecraft surface
6. Interface	$T_A(x_0) = T_B(x_0)$ and $-k_A\partial T_A/\partial x = -k_B\partial T_B/\partial x$	two layers of a composite wall

A steady one-dimensional problem (second-order equation) needs **two** boundary conditions; a transient problem also needs an **initial condition** $T(x, 0)$.

Does heat generation violate the first law? (2021 1b)

No. "Heat generation" means the *conversion* of another form of energy into thermal energy inside the solid: electrical energy in a resistance wire (I^2R), chemical energy in a reaction, nuclear energy in a fuel rod, absorbed radiation. Energy is not created; it changes form. The first law $E_{in} - E_{out} + E_{gen} = \Delta E_{stored}$ simply counts this conversion as a source term. If the source is counted too (e.g. the electrical work entering the wire), the total energy is conserved exactly.

Thermal contact resistance (2019 3a, 2022 2a)

When two solid surfaces are pressed together, they touch only at the peaks of their surface roughness; the gaps are filled with air (a poor conductor). Heat must squeeze through the few contact spots, so there is a sudden temperature drop ΔT at the interface. **Thermal contact resistance** $R_c = \Delta T_{interface}/(Q/A)$ (m^2K/W); its inverse is the thermal contact conductance h_c.

Factors: surface roughness; contact pressure (higher pressure → lower R_c); hardness of the materials; interfacial fluid (air, grease, helium); temperature; material conductivities; flatness. It is reduced by thermal grease or a soft metal foil (indium, aluminium).

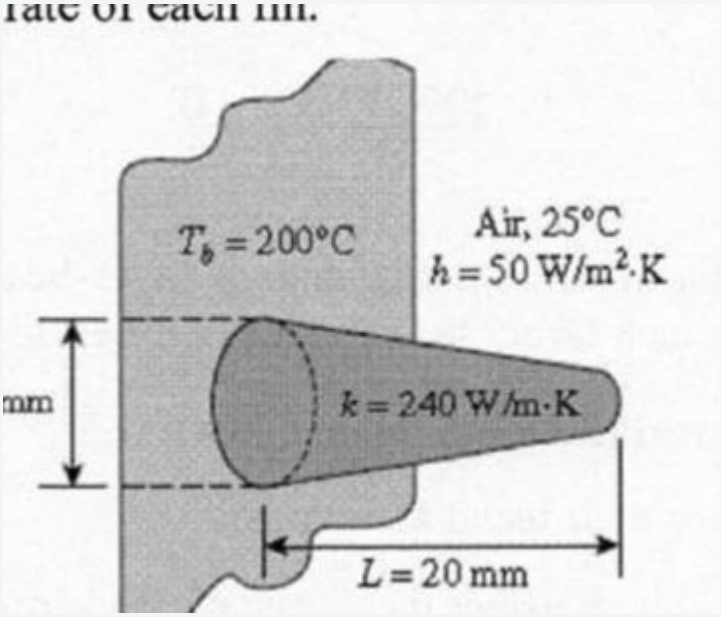
Memory trick: Modes: "**Fourier touches, Newton flows, Stefan shines**". Conduction equation: "**change in − change out + generation = storage**". Laplace = no source, no time; Poisson = source; Fourier/diffusion = time.

A4 Fins (extended surfaces): equation, solutions, efficiency, effectiveness, numericals

Pass set · Very likely · 100% · Medium

PAST-PAPER QUESTIONS

- **2023 Q3(a)** [12] What are fin effectiveness and efficiency? "Increasing fin length increases effectiveness but decreases efficiency": justify.
- **2023 Q3(b)** [8] Motor body 400 mm OD, 250 mm long, surface ≤ 62 °C when dissipating 350 W; longitudinal fins 18 mm thick, 38 mm high; h = 42, air 35 °C, k = 40. Find the number of fins.
- **2022 Q2(c)** [8] Explain (i) efficiency of fin, (ii) effectiveness of fin.
- **2021 Q2(a)** [5] Define heat transfer augmentation. Why is it necessary?
- **2021 Q2(c)** [15] Aluminium parabolic pin fins (blunt tip), L = 20 mm, D = 5 mm, k = 240, base 200 °C, air 25 °C, h = 50. Find efficiency and heat transfer per fin.



- **2020 Q2(c)** [15] Cylindrical fin, length L, insulated tip: derive the fin efficiency.
- **2020 Q1(b)** [15] Steel rod D = 2 cm, L = 25 cm, k = 50 W/m °C, h = 64 W/m^2°C, air 20 °C, base 120 °C. Find the heat loss.
- **2019 Q2(b)** [15] Derive the heat loss from a fin of uniform cross-section with negligible tip heat loss.

LEARN IT FROM ZERO

Fins, from zero

When convection is weak (small h), you cannot easily raise h, so you raise the **area**: stick thin metal plates or pins on the surface. Heat conducts along the fin and convects off its sides. But the fin gets cooler towards its tip, so the far end works less hard.

The fin equation $d^2\theta/dx^2 = m^2\theta$ with **$m = \sqrt{(hP/kA_c)}$** describes this. For a long fin or an insulated tip, $Q = \sqrt{(hPkA_c)}\ \theta_b \tanh(mL)$. **Efficiency** = actual heat / heat if the whole fin were at base temperature = $\tanh(mL)/(mL)$. **Effectiveness** = heat with fin / heat from the same base area without it (should be > 2).

Longer fin: more heat in total (effectiveness up) but the extra length is cooler, so efficiency drops. Beyond mL ≈ 2–3, extra length is wasted.

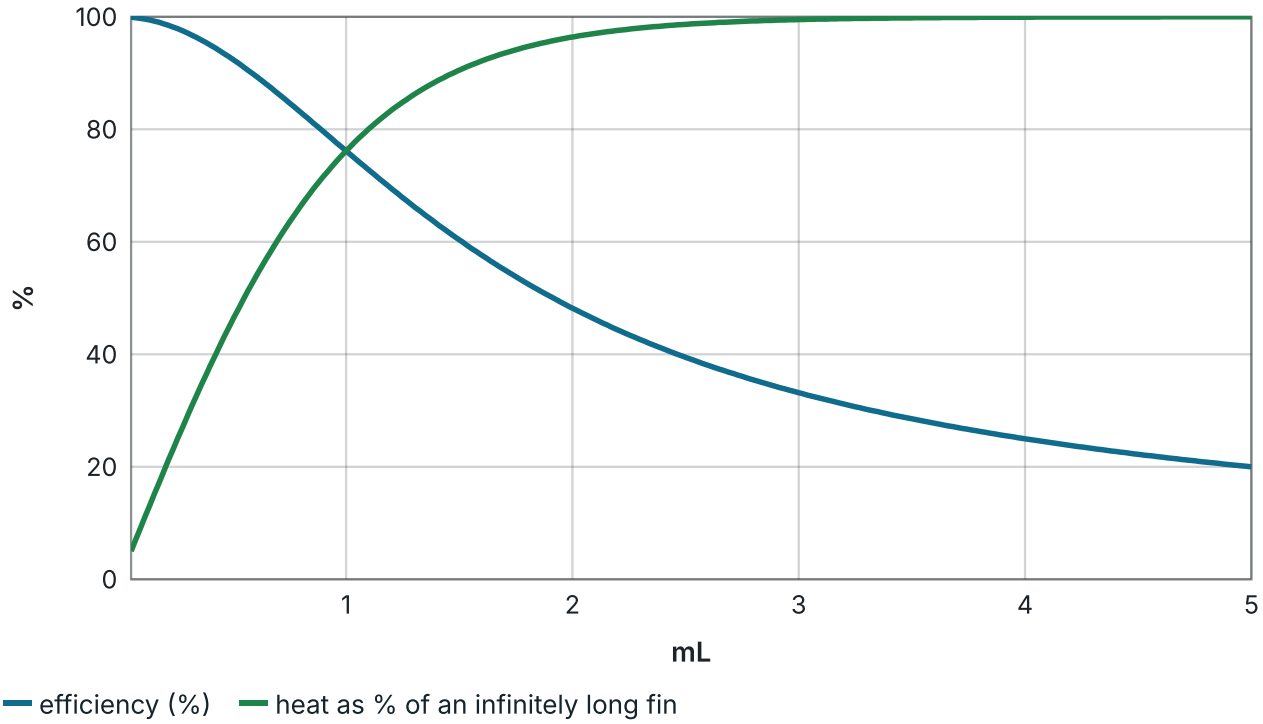
CALCULATOR Pin fin with insulated tip (2020 1(b) type)

Change any number; every step is recalculated and shown with the numbers put in.

1. Fin parameter: $m = \sqrt{(hP/kA_c)} = \sqrt{(4h/kD)} = \sqrt{(4 \times \mathbf{64} / (\mathbf{50} \times \mathbf{0.02}))} = \mathbf{16.000}$ 1/m
2. mL: $\mathbf{16} \times \mathbf{0.25} = \mathbf{4.000}$
3. $\sqrt{(hPkA_c)}$: $\sqrt{(\mathbf{64} \times \pi \times \mathbf{0.02} \times \mathbf{50} \times \pi \times \mathbf{0.02} \times \mathbf{0.02} / 4)} = \mathbf{0.2513}$ W/K
4. Heat loss: $Q = \sqrt{(hPkA_c)}\theta_b \tanh(mL) = \mathbf{0.2513} \times (\mathbf{120} - \mathbf{20}) \times \tanh(\mathbf{4}) = \mathbf{25.12}$ W
5. Fin efficiency: $\eta = \tanh(mL)/mL = \tanh(\mathbf{4}) / \mathbf{4} \times 100 = \mathbf{25.0}$ %
6. Fin effectiveness: $\varepsilon = Q/(hA_c\theta_b) = \mathbf{25.12} / (\mathbf{64} \times \pi \times \mathbf{0.02} \times \mathbf{0.02} / 4 \times (\mathbf{120} - \mathbf{20})) = \mathbf{12.5}$

INTERACTIVE Fin efficiency vs mL

Move the sliders and watch the curve change.

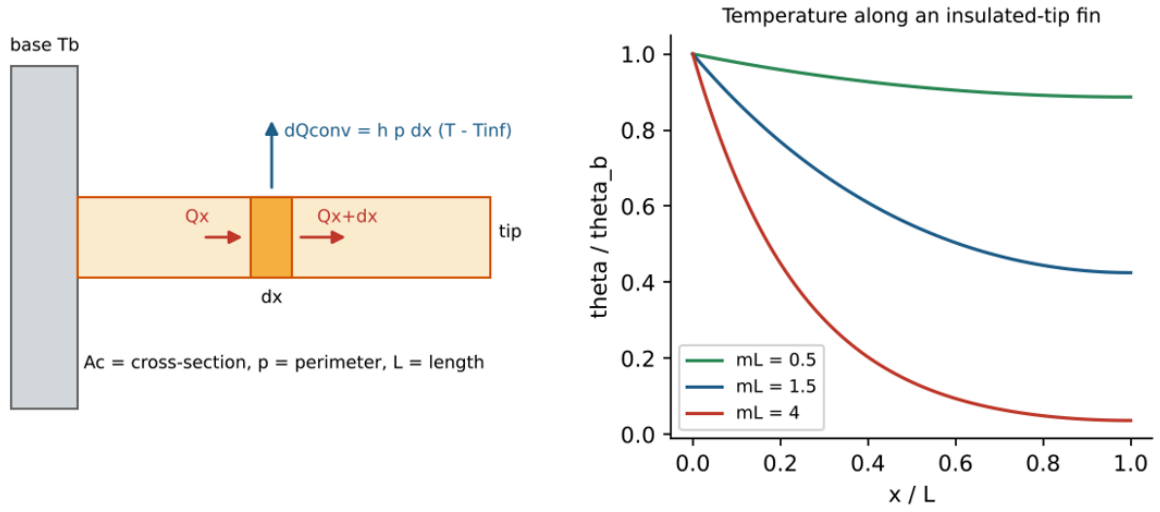


Design fins at mL around 1–1.5: about 76 % efficiency and 76–90 % of the maximum heat.

Before you start: what a fin does

Q = $hA_s\Delta T$: if h cannot be increased (air) and ΔT is fixed, the only way to remove more heat is to add surface area. A **fin** (extended surface) is a thin strip or pin of high-k metal attached to a surface. Examples: motorcycle engine cylinders, car radiators, transformer bodies, CPU heat sinks, air-conditioner coils.

Along the fin the temperature falls from T_b at the base towards T_∞, so the far parts are less effective. The fin equation describes this fall.



Fin element and temperature distribution

Heat transfer augmentation (2021 2a) means any technique that increases the heat transfer rate for the same temperature difference: fins, rough or ribbed surfaces, twisted tape inserts, vibration, higher flow velocity. It is needed to make heat exchangers, boilers, condensers and electronics cooling **smaller, lighter and cheaper**, to keep components below their safe temperature, and to save energy and material.

Derivation: fin equation and heat loss with an insulated tip (2019 2b, 2020 2c)

Side note: Assumptions: steady state, constant k, uniform h, no heat generation, one-dimensional conduction along x (thin fin), negligible radiation, uniform cross-section A_c with perimeter p.

Step (write this)	Why / what it means
Element dx: $Q_x = Q_{x+dx} + dQ_{conv}$, $dQ_{conv} = h(p\,dx)(T - T_\infty)$	Energy balance: conduction in = conduction out + convection from the side surface.
$Q_{x+dx} - Q_x = (dQ/dx)dx$, $Q = -kA_c dT/dx \Rightarrow kA_c d^2T/dx^2 - hp(T - T_\infty) = 0$	Taylor expansion + Fourier.
$\theta = T - T_\infty$, $m^2 = hp/(kA_c)$: $\mathbf{d^2\theta/dx^2 - m^2\theta = 0}$	Fin equation (T_∞ constant so dθ/dx = dT/dx).
General solution $\theta = C_1e^{mx} + C_2e^{-mx}$	Linear, second order, constant coefficients.
BCs: $\theta(0) = \theta_b$; insulated tip $d\theta/dx|_{x=L} = 0$	Tip heat loss negligible.
$C_1 + C_2 = \theta_b$; $C_1e^{mL} - C_2e^{-mL} = 0 \Rightarrow C_1 = \theta_b e^{-mL}/(2\cosh mL)$, $C_2 = \theta_b e^{mL}/(2\cosh mL)$	Solve the two equations.
$\mathbf{\theta/\theta_b = \cosh m(L - x)/\cosh mL}$	Temperature distribution.
$Q_{fin} = -kA_c d\theta/dx|_{x=0} = kA_c\theta_b m \tanh mL$	All the heat that leaves the fin enters at the base.

$Q_{fin} = \sqrt{(hpkA_c)}\ \theta_b \tanh(mL)$	Because $kA_cm = \sqrt{(hpkA_c)}$.

Other tips: infinitely long fin $Q = \sqrt{(hpkA_c)}\theta_b$; convective tip: use the insulated-tip formula with corrected length $L_c = L + A_c/p$ (= L + D/4 for a pin, L + t/2 for a plate).

Fin efficiency and effectiveness (2022 2c, 2023 3a, 2020 2c)

Step (write this)	Why / what it means
$n_{fin} = Q_{fin}/Q_{fin,max}$, $Q_{fin,max} = hA_{fin}\theta_b$	Efficiency: actual heat / heat if the whole fin were at base temperature (infinite k).
For a cylindrical pin with insulated tip $A_{fin} = pL$: $n = \sqrt{(hpkA_c)}\theta_b \tanh(mL)/(hpL\theta_b)$	
$\sqrt{(hpkA_c)}/(hp) = \sqrt{(kA_c/hp)} = 1/m$	Simplify.
$n_{fin} = \tanh(mL)/(mL)$, $m = \sqrt{(4h/kD)}$ for a pin	Always < 1; → 1 for short or very conductive fins.

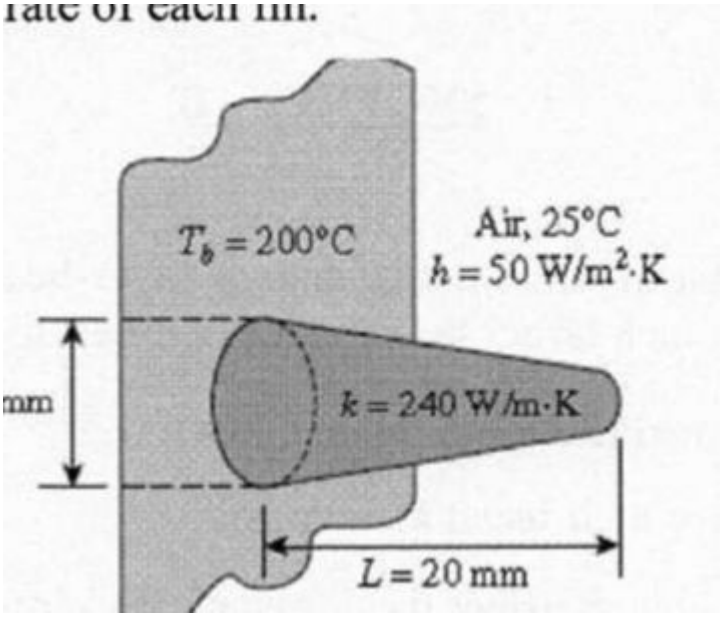
Effectiveness $\varepsilon_{fin} = Q_{fin}/(hA_b\theta_b)$: heat with the fin divided by heat that the base area A_b would lose without the fin. A fin is worth adding only if $\varepsilon > 2$. For a long fin $\varepsilon = \sqrt{(kp/hA_c)}$: high k, thin fins and low h (gases) make fins effective.

"Longer fin: more effective, less efficient" (2023 3a). $Q_{fin} \propto \tanh(mL)$, which keeps increasing with L (towards 1), so the heat removed and hence the effectiveness rise with length. But the efficiency tanh(mL)/mL falls because the added outer part is at a temperature close to T_∞ and contributes little: the area grows linearly while the heat grows less than linearly. Beyond mL ≈ 2-3 (tanh = 0.96-0.99), extra length adds weight and cost but almost no heat, so practical fins are designed with mL around 1-1.5.

Numerical 1: steel rod (2020 1b)

Step (write this)	Why / what it means
$m = \sqrt{(hp/kA_c)} = \sqrt{(4h/kD)} = \sqrt{(4 \times 64/(50 \times 0.02))} = 16\ m^{-1}$; $mL = 16 \times 0.25 = 4.0$	For a circular rod $p/A_c = 4/D$.
$\sqrt{(hpkA_c)} = \sqrt{(64 \times \pi \times 0.02 \times 50 \times \pi \times 0.02^2/4)} = 0.2513$ W/K	
$Q = 0.2513 \times (120 - 20) \times \tanh 4 = 25.1$ W	tanh 4 = 0.9993: the rod behaves as an infinitely long fin; including tip convection ($L_c = L + D/4$) also gives 25.1 W.

Numerical 2: parabolic pin fin with blunt tip (2021 2c)



Question figure (2021 paper)

Step (write this)	Why / what it means
$m = \sqrt{(4h/kD)} = \sqrt{(4 \times 50/(240 \times 0.005))} = 12.91\ m^{-1}$; $mL = 12.91 \times 0.02 = 0.258$	Same m definition as a pin.
From Cengel Table 3-3 (parabolic pin, blunt tip): $n = 3I_1(4mL/3)/[2mL\ I_0(4mL/3)]$	I_0, I_1 = modified Bessel functions (table in Cengel).
$4mL/3 = 0.344$; $I_1 = 0.1747$, $I_0 = 1.0298 \Rightarrow$ **$n = 3 \times 0.1747/(2 \times 0.258 \times 1.0298) = 0.985$**	Short, highly conductive fin: almost isothermal.
$A_{fin} = (\pi D^4/96L^2)\{\{[16(L/D)^2 + 1]^{3/2} - 1\}\} = 2.106\times10^{-4}\ m^2$	Surface area of the parabolic pin (L/D = 4).
$Q = nhA_{fin}(T_b - T_\infty) = 0.985 \times 50 \times 2.106\times10^{-4} \times 175 = 1.82$ W per fin	

Numerical 3: motor body with longitudinal fins (2023 3b)

Step (write this)	Why / what it means
$\theta_b = 62 - 35 = 27\ °C$	Allowed surface excess temperature.
Bare motor surface: $Q_{bare} = h\pi DL\theta_b = 42 \times \pi \times 0.4 \times 0.25 \times 27 = 356$ W	First check if fins are needed at all.
Each fin (length 0.25 m along the motor, thickness t = 0.018, height H = 0.038): $p = 2(0.25 + 0.018) = 0.536$ m, $A_c = 0.25 \times 0.018 = 0.0045\ m^2$, $m = \sqrt{(42 \times 0.536/(40 \times 0.0045))} = 11.18\ m^{-1}$	Fin cross-section seen by the heat flowing outwards.
$Q_{fin} = \sqrt{(hpkA_c)}\theta_b \tanh(mH) = 2.016 \times 27 \times \tanh(0.425) = 21.8$ W (26.2 W with tip convection)	Insulated-tip formula.
n fins: $Q = n\ Q_{fin} + h(\pi D - nt)L\theta_b = 356 + n(21.8 - 5.1) \geq 350$ W	Fins replace part of the bare surface.
Result: the bare surface already dissipates 356 W > 350 W, so theoretically no fin is needed (n	With the printed data the requirement is met without fins.

= 0).	

Flag: with the printed numbers the bare motor surface (end faces ignored) already removes 356 W at 62 °C. If the examiner intended a larger heat load or a lower h, the method above gives n directly: $n = (Q_{required} - 356)/(Q_{fin} - htL\theta_b) = (Q_{required} - 356)/16.7$. For example, 500 W would need (500 − 356)/16.7 = 8.6 → **9 fins**. Write the method fully and state the conclusion.

Memory trick: "**m = √(hp/kA)**, Q = √(hpkA)θ tanh mL, η = tanh mL/mL". Long fin: "effective but not efficient".

WRITE IN EXAM **Write this in the exam (this much is enough for full marks)**

2020 1(b) Given: D = 0.02 m, L = 0.25 m, k = 50 W/m°C, h = 64 W/m²°C, T_b = 120 °C, T_∞ = 20 °C

Fin parameter: $m = \sqrt{(4h/kD)}$

$= \sqrt{(4 \times 64/(50 \times 0.02))}$

$= \boxed{\mathbf{16\ m^{-1};\ mL = 4.0}}$

$\sqrt{(hPkA_c)}$: $\sqrt{(64 \times \pi \times 0.02 \times 50 \times \pi \times 0.02^2/4)}$

$= \sqrt{(0.0632)}$

$= \boxed{\mathbf{0.2513\ W/K}}$

Heat loss: $Q = \sqrt{(hPkA_c)}(T_b - T_\infty) \tanh(mL)$

$= 0.2513 \times 100 \times 0.9993$

$= \boxed{\mathbf{25.1\ W}}$

tanh 4 ≈ 1, so the rod acts as an infinitely long fin.

A3 Critical radius (thickness) of insulation Pass set Very likely · 100% Medium

PAST-PAPER QUESTIONS

- **2023 Q2(b)** [12] Significance of critical thickness of insulation; derive it for a cylindrical object.
- **2022 Q2(b)** [10] Same question: insulation outer radius below critical, insulation removed; justify mathematically.
- **2022 Q2(a)** [5] Define critical thickness of insulation.
- **2021 Q2(b)** [15] Explain the concept of critical thickness of insulation. Derive the critical radius for a small hollow pipe.
- **2020 Q2(a)** [10] Define critical radius of insulation. If the outer radius of insulation is less than critical and the insulation is removed, will heat transfer increase or decrease? Justify.
- **2019 Q3(a)** [3] Define and explain critical thickness of insulation.

LEARN IT FROM ZERO

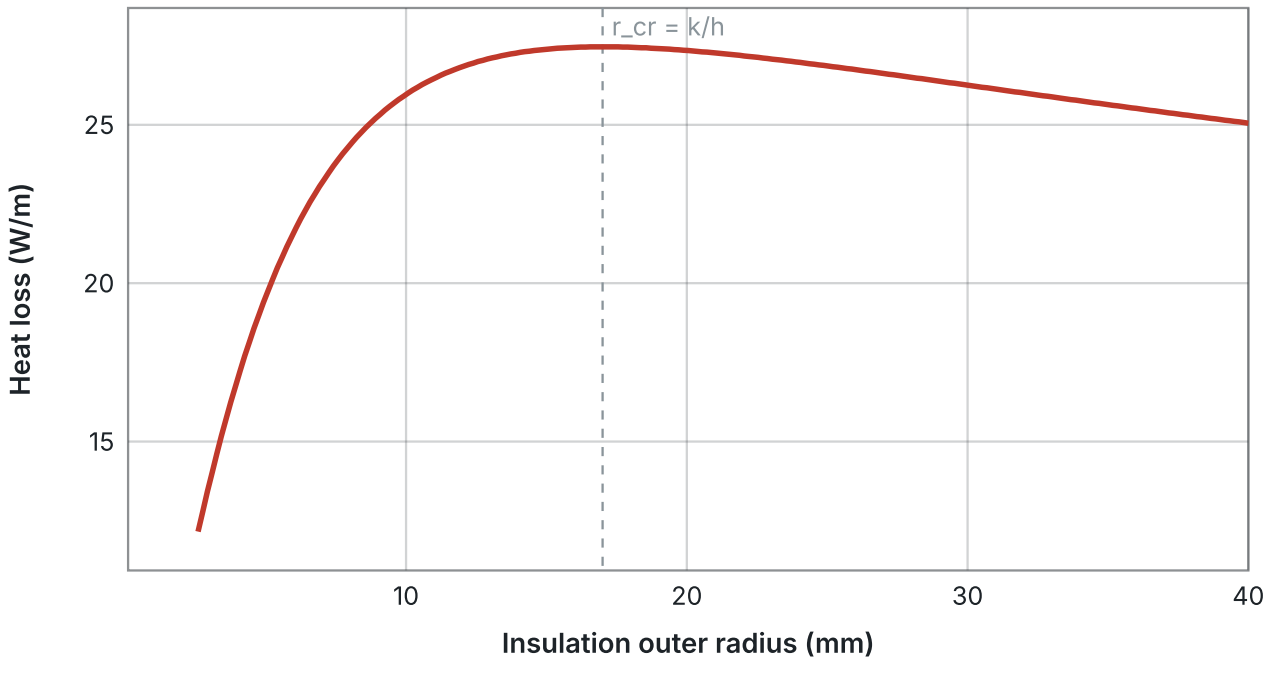
Critical radius of insulation, from zero

Wrapping insulation round a thin pipe does two things at once. It adds **conduction resistance** (good, less heat loss) but it also makes the outer surface **bigger**, so there is more area for convection to the air (bad, more heat loss). For a thin wire the second effect wins at first: adding insulation *increases* heat loss, until the outer radius reaches the **critical radius r_{cr} = k/h** (cylinder) or 2k/h (sphere). Beyond that, more insulation reduces the loss.

So if the insulation's outer radius is below r_{cr} and you remove it, heat loss goes **down**. This is used on purpose for electric cables: a little insulation helps them cool.

INTERACTIVE **Heat loss vs insulation outer radius**

Move the sliders and watch the curve change.

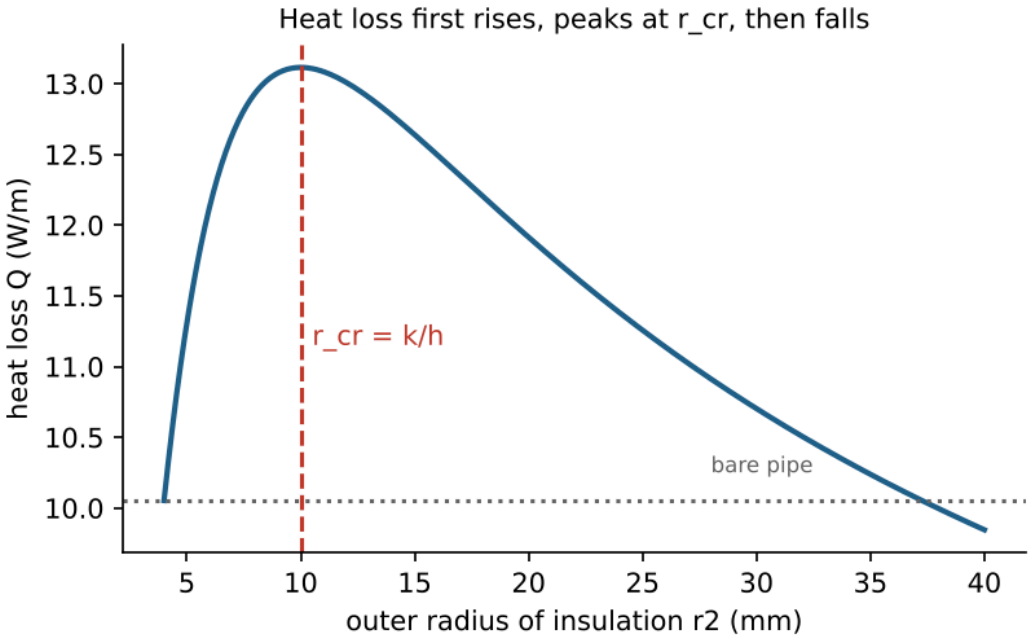


Critical radius r_{cr} = k/h: **17.00** mm

The insulation term uses radii in mm (ratio) and the convection term converts to metres. Heat loss peaks at r_{cr}.

Before you start: why insulation can increase heat loss

Wrapping insulation around a pipe adds conduction resistance $\ln(r_2/r_1)/(2\pi kL)$, which *reduces* heat loss, but it also enlarges the outer surface, so the convection resistance $1/(h2\pi r_2 L)$ *falls*, which *increases* heat loss. For a thin pipe the second effect wins at first.



Heat loss from an insulated pipe as the insulation radius grows

Definition and significance (2019 3a, 2020 2a, 2022 2a, 2023 2b)

The **critical radius of insulation** r_{cr} is the outer radius of insulation at which the heat loss from a pipe (or sphere) is **maximum**. Up to r_{cr} adding insulation increases heat loss; beyond it, more insulation reduces heat loss. The **critical thickness** is $r_{cr} - r_1$.

- Thermal insulation of a pipe is effective only if the outer radius exceeds r_{cr}; for steam pipes r_{cr} (a few mm) is far smaller than the pipe, so insulation always helps.
- Electric wires and cables: plastic insulation thinner than r_{cr} actually **increases** heat dissipation, so the wire runs cooler and can carry more current.
- Small-diameter tubes (refrigerant lines, thin probes) must be insulated beyond r_{cr}, or the insulation is counter-productive.

Derivation for a cylinder (2021 2b, 2023 2b)

Step (write this)	Why / what it means
Pipe outer radius r_1 at T_1; insulation k, outer radius r_2; air h, T_∞; length L	Set-up.
$Q = (T_1 - T_\infty)/[\ln(r_2/r_1)/(2\pi kL) + 1/(h2\pi r_2 L)]$	Insulation and convection resistances in series.

Q is maximum when the denominator $R_{tot}(r_2)$ is minimum: $dR_{tot}/dr_2 = 0$	Numerator is fixed.
$dR_{tot}/dr_2 = 1/(2\pi k L r_2) - 1/(2\pi h L r_2^2) = 0$	Differentiate ln r_2 and $1/r_2$.
$\mathbf{r_{cr,cylinder} = k/h}$	Second derivative is positive, so it is a minimum of R (maximum of Q).
Sphere: $R = (r_2 - r_1)/(4\pi k\, r_1 r_2) + 1/(4\pi h r_2^2) \Rightarrow$ $\mathbf{r_{cr,sphere} = 2k/h}$	Same procedure.

Example: glass wool k = 0.05 W/m K in still air h = 5 W/m^{2}K gives r_{cr} = 1 cm.

Insulation thinner than critical is removed: heat loss rises or falls? (2020 2a, 2022 2b)

Heat transfer decreases. Mathematically: for $r_1 \le r_2 < r_{cr}$, $dR_{tot}/dr_2 = (1/2\pi L r_2)(1/k - 1/(hr_2)) < 0$ because $hr_2 < k$. So R_{tot} *falls* as r_2 grows from r_1 to r_2, i.e. $R_{tot}(r_2) < R_{tot}(r_1) = 1/(h2\pi r_1 L)$ (bare pipe). Since $Q = \Delta T/R_{tot}$, the insulated pipe loses **more** heat than the bare pipe; removing the insulation increases the resistance and the heat loss **decreases** (for the same pipe surface temperature).

> **Memory trick:** "**Cylinder k over h, sphere two k over h**". Below r_{cr} insulation acts like a fin; above it, like a blanket.

A5 Transient conduction: lumped system, Heisler/one-term solution, semi-infinite solid

Next to add · Very likely · 87% · Medium

PAST-PAPER QUESTIONS

- **2023 Q3(c)** [15] Same shaft problem (35 cm, 400 °C → 150 °C surroundings, h = 60, 20 min).
- **2022 Q3(a)** [8] What is a lumped system? Copper ball D = 12 cm, k = 401, h = 15: can it be treated as lumped?
- **2022 Q3(b)** [14] For a lumped system show $\theta/\theta_i = e^{-Bi \cdot Fo}$.
- **2022 Q3(c)** [13] Brass plates 3 cm thick (k = 110, ρ = 8530, c_p = 380, α = 33.9×10^{-6}) at 25 °C, oven 700 °C for 10 min, h = 80. Find surface temperature.
- **2021 Q3(a)** [17] Semi-infinite solid at T_i, surface suddenly at T_s: obtain $(T - T_s)/(T_i - T_s) = 1 - \text{erfc}(\eta)$ using the similarity variable.
- **2021 Q3(b)** [10] Two semi-infinite solids A and B brought into contact: show $T_s = (T_{A,i} + \alpha T_{B,i})/(1 + \alpha)$, $\alpha = \sqrt{(k\rho c_p)_B}/\sqrt{(k\rho c_p)_A}$.
- **2021 Q3(c)** [8] Snowpack at −10 °C for 60 days, soil k = 0.4, α = 0.15×10^{-6}, initially 15 °C: minimum burial depth of water pipes.
- **2019 Q3(b)** [15] For a lumped system show $\theta/\theta_i = e^{-Bi \cdot Fo}$.
- **2019 Q3(c)** [14] Stainless steel shaft D = 35 cm (k = 14.9, ρ = 7900, c_p = 477, α = 3.95×10^{-6}) from 400 °C cooled in 150 °C with h = 60. Centre temperature after 20 min and heat transfer per m.

LEARN IT FROM ZERO

Transient conduction, from zero

Drop a hot metal ball into water. Does its inside stay at one temperature while it cools? Only if heat moves through the ball much more easily than it leaves the surface. The **Biot number Bi = hL_c/k** compares the two ($L_c = V/A_s$). If Bi ≤ 0.1 the body is "lumped": one temperature everywhere, and it cools exponentially: **$(T - T_\infty)/(T_i - T_\infty) = e^{-bt}$**, $b = hA_s/(\rho V c_p)$.

If Bi > 0.1 the centre lags behind the surface and you need the Heisler charts or the one-term series solution, using the **Fourier number Fo = $\alpha t/L^2$** (dimensionless time).

CALCULATOR **Lumped-system check and temperature after time t**

Change any number; every step is recalculated and shown with the numbers put in.

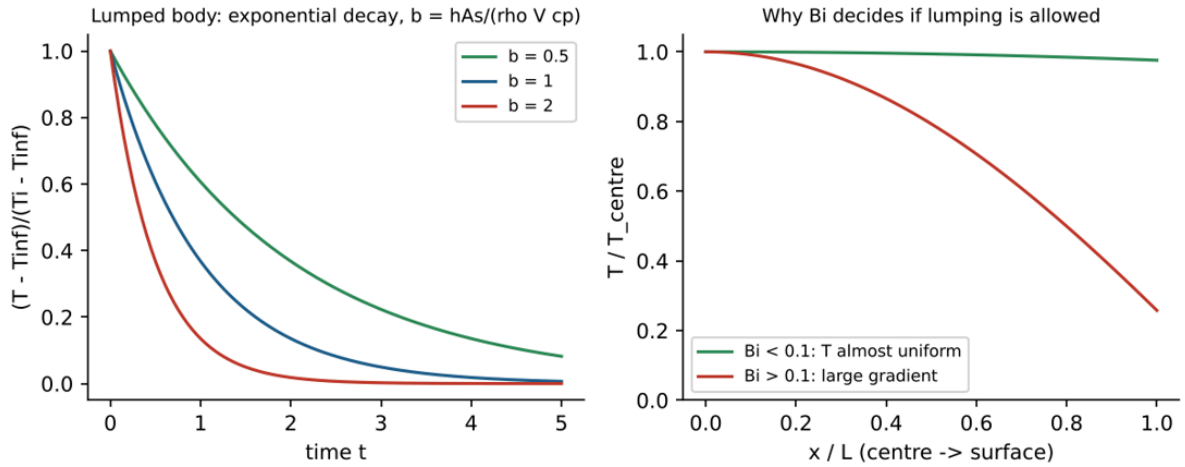
1. Characteristic length V/A_s: **1** == 1 ? **0.12** / 6 : **1** == 2 ? **0.12** / 4 : **0.12** / 2 = **0.0200** m
2. Biot number: $Bi = hL_c/k$ = **15** × **0.02** / **401** = **7.481 × 10^{-4}**
3. Lumped analysis: **valid (Bi ≤ 0.1)**
4. $b = h/(\rho c_p L_c)$: **15** / (**8933** × **385** × **0.02**) = **2.181 × 10^{-4}** 1/s
5. Temperature after t: $T = T_\infty + (T_i - T_\infty)e^{-bt}$ = **25** + (**200** - **25**) × exp(-**2.181 × 10^{-4}** × **600**) = **178.54** °C

6. Time constant 1/b: 1 / **2.181×10^{-4}** = **4586** s

Before you start: how a body heats up or cools down

When a hot object is put in cold air, heat must travel from its inside to its surface (conduction, resistance L/k) and then from the surface to the air (convection, resistance 1/h). The **Biot number** Bi = hL_c/k compares these two resistances. If Bi < 0.1 the inside resistance is negligible and the whole body has nearly one temperature: a **lumped system**.

Fourier number Fo = τ = $\alpha t/L_c^2$ is dimensionless time: how far heat has penetrated. $L_c = V/A_s$ (sphere D/6, long cylinder D/4, plate half-thickness).



Lumped cooling curves and the meaning of the Biot number

Lumped system: definition and check (2022 3a)

A **lumped system** is a body whose interior temperature remains essentially uniform at all times during a heat transfer process, so T = T(t) only. Valid when Bi = $hL_c/k \le 0.1$.

Step (write this)	Why / what it means
$L_c = V/A_s = (\pi D^3/6)/(\pi D^2) = D/6 = 0.12/6 = 0.02$ m	Characteristic length of a sphere.
Bi = 15 × 0.02/401 = 0.00075 < 0.1 ⇒ lumped analysis is appropriate	Copper conducts so well that its inside stays uniform.

Derivation: $\theta/\theta_i = e^{-Bi \cdot Fo}$ (2019 3b, 2022 3b)

Step (write this)	Why / what it means
Body: volume V, surface A_s, density ρ, c_p, initially T_i, put in fluid T_∞ with h	Set-up.
$hA_s(T_\infty - T)dt = \rho V c_p dT$	Heat in by convection during dt = increase of stored energy.
$\theta = T - T_\infty$, $d\theta = dT$: $d\theta/\theta = -(hA_s/\rho V c_p)dt$	Separate variables.
$\ln(\theta/\theta_i) = -bt$, $b = hA_s/(\rho V c_p)$	Integrate from t = 0 (θ_i) to t.
$bt = hA_s t/(\rho V c_p) = (hL_c/k)(kt/\rho c_p L_c^2) = Bi \cdot Fo$	Multiply and divide by kL_c; $L_c = V/A_s$, $\alpha = k/\rho c_p$.

$\theta/\theta_i = (T - T_\infty)/(T_i - T_\infty) = e^{-bt} = e^{-Bi \cdot Fo}$	Exponential approach to the fluid temperature; 1/b is the time constant.

Numerical: brass plates in an oven (2022 3c)

The printed thickness "3 m" is corrected by hand on the paper to **3 cm** (also shown in the figure).

Step (write this)	Why / what it means
$L_c = V/A_s$ = half thickness = 0.015 m (heated on both faces)	Large plate: A_s = 2 × face area.
$Bi = hL_c/k = 80 \times 0.015/110 = 0.011 < 0.1 \Rightarrow$ lumped	Check first.
$b = h/(\rho c_p L_c) = 80/(8530 \times 380 \times 0.015) = 0.001645\ s^{-1}$	
$(T - 700)/(25 - 700) = e^{-0.001645 \times 600} = e^{-0.987} = 0.373$	
$T = 700 - 675 \times 0.373 = 448$ °C	Surface temperature after 10 min (one-term exact solution also gives 448 °C).

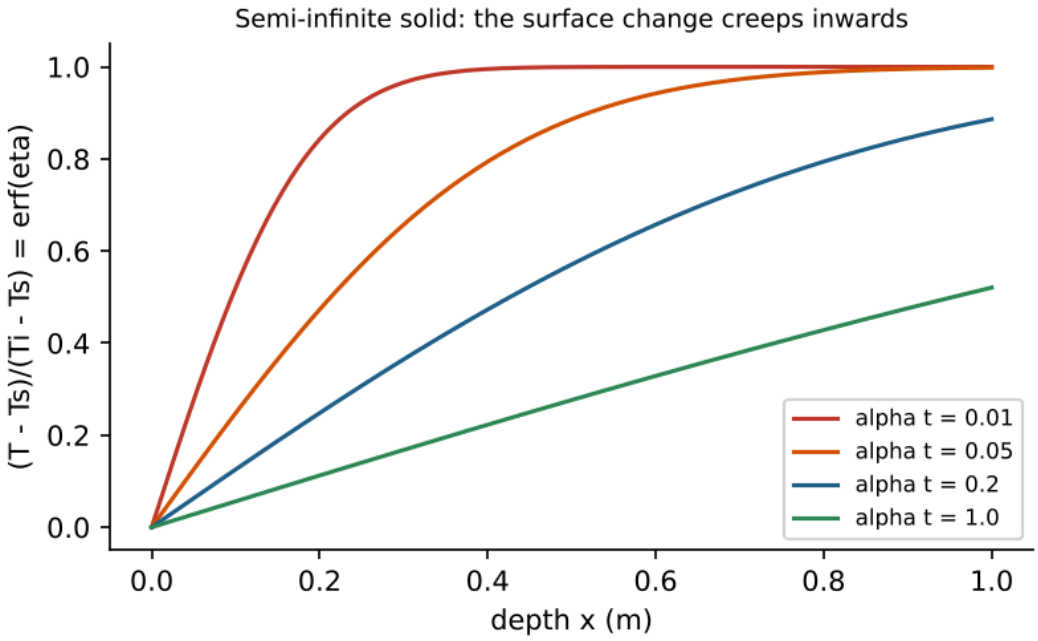
Numerical: long stainless-steel shaft (2019 3c, 2023 3c)

Before you start: one-term (Heisler) solution

When Bi > 0.1 the centre and surface differ, and we use the exact series solution, keeping only its first term (valid for $\tau > 0.2$): $\theta_0 = (T_0 - T_\infty)/(T_i - T_\infty) = A_1 e^{-\lambda_1^2 \tau}$, with λ_1, A_1 read from Cengel Table 4-2 against $Bi = hr_0/k$. The Heisler charts are graphs of the same formula. Heat transferred: $Q/Q_{max} = 1 - 2\theta_0 J_1(\lambda_1)/\lambda_1$ (cylinder), $Q_{max} = \rho V c_p (T_i - T_\infty)$.

Step (write this)	Why / what it means
r_0 = 0.175 m; $Bi = hr_0/k = 60 \times 0.175/14.9 = 0.705$	Here Bi uses r_0, not V/A.
Table 4-2: $\lambda_1 = 1.0904$, $A_1 = 1.1548$	Interpolated for Bi = 0.705.
$\tau = \alpha t/r_0^2 = 3.95 \times 10^{-6} \times 1200/0.175^2 = 0.155$	Slightly below 0.2, so the one-term answer is a close approximation (Cengel accepts it).
$\theta_0 = 1.1548\ e^{-(1.0904)^2(0.155)} = 0.961$	
$T_0 = 150 + 0.961(400 - 150) = 390$ °C	The centre has hardly cooled after 20 min (low k, large shaft).
$Q_{max} = \rho \pi r_0^2 L c_p (T_i - T_\infty) = 7900 \times \pi \times 0.175^2 \times 1 \times 477 \times 250 = 90.6$ MJ per m	
$Q/Q_{max} = 1 - 2(0.961)J_1(1.0904)/1.0904 = 1 - 2 \times 0.961 \times 0.4679/1.0904 = 0.175$	$J_1(1.09) = 0.468$ from the Bessel table.
$Q = 0.175 \times 90.6 = 15.9$ MJ per metre of shaft	Cengel Example 4-5 gives 390 °C and 15,960 kJ (chart readings).

Semi-infinite solid: derivation with the similarity variable (2021 3a)



Temperature profiles in a semi-infinite solid at increasing times

Step (write this)	Why / what it means
$\partial^2T/\partial x^2 = (1/\alpha)\partial T/\partial t$; $T(x,0) = T_i$, $T(0,t) = T_s$, $T(\infty,t) = T_i$	Semi-infinite body: deep enough that the far end never feels the change.
Define $n = x/\sqrt{(4\alpha t)}$	Similarity variable: combines x and t into one.
$\partial T/\partial t = (dT/dn)(\partial n/\partial t) = -(x/2t\sqrt{(4\alpha t)})dT/dn$; $\partial^2T/\partial x^2 = (1/4\alpha t)d^2T/dn^2$	Chain rule.
Substitute: $\mathbf{d^2T/dn^2 = -2n\ dT/dn}$	Partial differential equation → ordinary one.
BCs: $n = 0$ ($x = 0$): $T = T_s$; $n \to \infty$ ($x \to \infty$ or $t = 0$): $T = T_i$	Initial and far conditions merge into one.
Let $w = dT/dn$: $dw/w = -2ndn \Rightarrow w = C_1e^{-n^2} \Rightarrow T = C_1\int_0^n e^{-u^2}du + C_2$	Integrate twice.
$n = 0$: $C_2 = T_s$; $n \to \infty$: $\int_0^\infty e^{-u^2}du = \sqrt{\pi}/2 \Rightarrow C_1 = 2(T_i - T_s)/\sqrt{\pi}$	
$\mathbf{(T - T_s)/(T_i - T_s) = (2/\sqrt{\pi})\int_0^n e^{-u^2}du = erf(n) = 1 - erfc(n)}$	erf = error function (tabulated).

Contact of two semi-infinite solids (2021 3b)

Step (write this)	Why / what it means
Both bodies are semi-infinite; at contact the interface instantly takes a temperature T_s that stays constant	Known result for two semi-infinite solids.
Heat flux into a semi-infinite solid with surface step: $q_s = k(T_s - T_i)/\sqrt{(\pi\alpha t)}$	Differentiate the erf solution at x = 0.
Heat leaving A = heat entering B: $k_A(T_{A,i} - T_s)/\sqrt{(\pi\alpha_A t)} = k_B(T_s - T_{B,i})/\sqrt{(\pi\alpha_B t)}$	Energy balance at the interface (no storage there).

$k/\sqrt{\alpha} = k/\sqrt{(k/\rho c_p)} = \sqrt{(k\rho c_p)}$: $\sqrt{(k\rho c_p)_A}(T_{A,i} - T_s) = \sqrt{(k\rho c_p)_B}(T_s - T_{B,i})$	$\sqrt{t}$ and $\sqrt{\pi}$ cancel: T_s does not change with time.
$T_s = (T_{A,i} + aT_{B,i})/(1 + a)$, $a = \sqrt{(k\rho c_p)_B}/\sqrt{(k\rho c_p)_A}$	Divide by $\sqrt{(k\rho c_p)_A}$. This explains why metal "feels colder" than wood at the same temperature.

Numerical: burial depth of water pipes (2021 3c)

Step (write this)	Why / what it means
$(T - T_s)/(T_i - T_s) = (0 - (-10))/(15 - (-10)) = 0.40 = \text{erf}(n)$	Pipe must stay at or above 0 °C.
From the erf table: $n = 0.37$	erf(0.37) = 0.40.
$t = 60 \times 24 \times 3600 = 5.184\times10^6$ s; $\sqrt{(\alpha t)} = \sqrt{(0.15\times10^{-6} \times 5.184\times10^6)} = 0.882$ m	
$x = 2n\sqrt{(\alpha t)} = 2 \times 0.37 \times 0.882 = 0.65$ m	Bury the pipes at least 65 cm deep (Cengel Example 4-9).

Memory trick: "Bi below 0.1 → lump it"; $e^{-Bi\cdot Fo}$. Shaft: $Bi = hr_0/k$ → table λ_1, A_1 → $\theta_0 = A_1e^{-\lambda^2\tau}$. Ground: erf(η), $\eta = x/2\sqrt{(\alpha t)}$.

WRITE IN EXAM **Write this in the exam (this much is enough for full marks)**

2022 3(a) Copper ball D = 12 cm, k = 401 W/m K, h = 15 W/m²K

Characteristic length: $L_c = V/A_s = D/6$

$= 0.12/6$

$=$ **0.02 m**

Biot number: $Bi = hL_c/k$

$= 15 \times 0.02/401$

$=$ **0.00075**

Bi = 0.00075 < 0.1, so the ball **can** be treated as a lumped system (its inside stays at one temperature).

Section B: Convection and mass transfer

B1 Convection fundamentals: mechanism, boundary layers, dimensionless numbers, film temperature

Pass set · Very likely · 100% · Easy

PAST-PAPER QUESTIONS

- **2023 Q5(a)** [10] With a sketch, write down the physical mechanism of convection heat transfer.
- **2023 Q5(b)** [12] Differentiate hydrodynamic and thermal boundary layers; justify that h decreases as δ_t grows.
- **2023 Q6(a)** [8] Significance of (i) Prandtl, (ii) Rayleigh, (iii) Nusselt numbers.
- **2022 Q5(a)** [11] Explain the mechanism of convection. What are (i) hydrodynamic and (ii) thermal boundary layers?
- **2022 Q5(b)** [12] The thermal boundary layer thickens in the flow direction but h decreases. Why?
- **2022 Q6(b)** [6] Define film temperature and its significance.
- **2022 Q8(b)** [10] Methods of evaluating the convective coefficient; the analogy between heat and momentum transfer and the numerical method.
- **2021 Q5(a)** [13] Explain turbulent boundary-layer heat transfer. Why is the turbulent h much larger than laminar?
- **2021 Q6(b)** [4] Significance of Grashof number.
- **2021 Q6(a)** [16] Using dimensional analysis show Nu = f(Re, Pr) for forced convection over a flat plate.
- **2020 Q4(a)** [10] Physical significance of Nusselt and Prandtl numbers.
- **2020 Q6(b)** [8] Physical significance of Grashof number; how does it differ from Reynolds number?
- **2019 Q5(a)** [5] Distinguish between (i) thermal and hydrodynamic boundary layer, (ii) developing and developed flow.
- **2019 Q6(b)** [12] Explain turbulent boundary-layer heat transfer. Why is the turbulent h much larger than laminar?
- **2019 Q6(a)** [6] Significance of Nusselt and Prandtl numbers.

★ REMEMBER IT

- Dimensionless numbers: **Re** = inertia/viscous (laminar or turbulent?), **Pr** = ν/α (momentum vs heat spreading; δ/δ_t), **Nu** = hL/k (how much convection beats pure conduction), **Gr** = buoyancy/viscous (natural convection's Re), **Ra** = Gr·Pr, **St** = Nu/(Re Pr).
- h falls along a plate because the thermal boundary layer thickens: the same ΔT now spreads over a thicker layer, so the gradient at the wall (and $h = -k(\partial T/\partial y)_{wall}/\Delta T$) gets smaller.
- Film temperature $T_f = (T_s + T_\infty)/2$: evaluate air properties here.

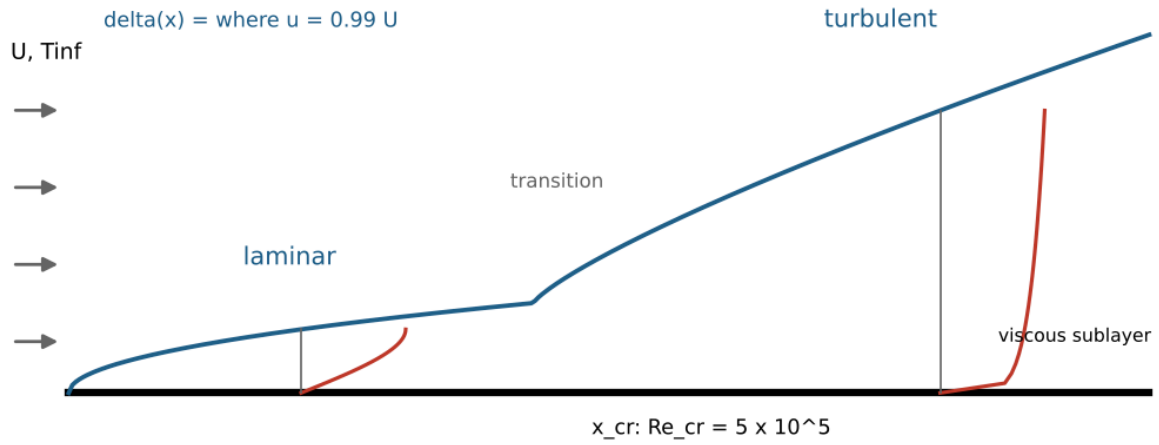
Before you start: what convection is

Convection is heat transfer between a surface and a moving fluid. Right at the wall the fluid is at rest (no-slip), so heat enters the fluid by **conduction**; the moving fluid then **carries it away** (advection) and brings fresh cold fluid to the wall. Faster flow = steeper temperature gradient at the wall = more heat.

Newton's law of cooling $Q = hA_s(T_s - T_\infty)$. The whole of convection theory is about finding h.

Since the wall layer is conducting: $h = -k_{fluid}(\partial T/\partial y)_{y=0}/(T_s - T_\infty)$.

Physical mechanism of convection (2022 5a, 2023 5a)



Velocity and thermal boundary layers on a heated flat plate

1. A cold fluid at T_∞ flows over a hot surface at T_s.
2. The fluid layer touching the wall sticks to it (velocity 0) and takes the wall temperature; heat passes into this layer by pure conduction: $q = -k(\partial T/\partial y)_{y=0}$.
3. The heated fluid is swept downstream by the flow (energy carried by bulk motion) and replaced by cooler fluid, so the temperature gradient at the wall stays steep.
4. Convection is therefore **conduction + fluid motion**. It is **forced** if the motion is caused by a fan or pump, **natural (free)** if it is caused by buoyancy (density differences from heating).
5. h depends on: fluid properties (k, μ, ρ, c_p), velocity, geometry and surface roughness, and the flow type (laminar/turbulent). Typical h: free convection of gases 2-25, forced gases 25-250, forced liquids 50-20,000, boiling/condensation 2500-100,000 W/m^2K.

Hydrodynamic and thermal boundary layers (2022 5a, 2019 5a, 2023 5b)

Hydrodynamic (velocity) boundary layer	Thermal boundary layer
Region near the wall where velocity changes from 0 (wall) to $0.99U_\infty$	Region where temperature changes from T_s to $T_s + 0.99(T_\infty - T_s)$
Thickness δ, caused by viscosity (momentum diffusion ν)	Thickness δ_t, caused by heat diffusion α
Gives wall shear stress $\tau_s = \mu(\partial u/\partial y)_0$ and friction	Gives wall heat flux $q_s = -k(\partial T/\partial y)_0$ and h
Exists whenever a fluid flows over a surface	Exists only if $T_s \neq T_\infty$

Their relative thickness is set by the Prandtl number: $\delta/\delta_t \approx Pr^{1/3}$. Gases ($Pr \approx 0.7$): $\delta \approx \delta_t$; oils ($Pr >> 1$): $\delta_t << \delta$; liquid metals ($Pr << 1$): $\delta_t >> \delta$.

Developing vs fully developed flow (2019 5a): at a pipe entrance the boundary layers grow from the wall; this is the **developing (entrance)** region, where the velocity/temperature profile changes with x and h is high. Once the boundary layers meet at the centre the profile stops changing shape with x: **fully developed** flow, with constant friction factor and constant h. (Hydrodynamic entry length $L_h \approx 0.05ReD$ laminar, $\approx 10D$ turbulent.)

Why h decreases as the thermal boundary layer grows (2022 5b, 2023 5b)

Step (write this)	Why / what it means
$h = -k(\partial T/\partial y)_{y=0}/(T_s - T_\infty)$	Definition of h from the wall conduction.
The whole temperature drop $T_s - T_\infty$ takes place across $\delta_t \Rightarrow (\partial T/\partial y)_0 \approx (T_\infty - T_s)/\delta_t$	Rough estimate of the wall gradient.
$h \approx k/\delta_t$	The film acts like a conducting layer of thickness δ_t.
Laminar flat plate: $\delta_t \propto x^{1/2} \Rightarrow h_x \propto x^{-1/2}$	As the layer thickens downstream, the gradient at the wall becomes gentler, so h falls. Heat must "conduct" through a thicker insulating fluid layer.

Turbulent boundary layer heat transfer (2019 6b, 2021 5a)

Beyond $Re_x \approx 5\times10^5$ the laminar boundary layer becomes unstable and turns turbulent. The turbulent layer has three zones: a thin **viscous (laminar) sublayer** at the wall, a **buffer layer**, and the **turbulent core** full of random eddies.

- Eddies move lumps of fluid across the layer, carrying hot fluid outwards and cold fluid towards the wall. This **eddy (turbulent) mixing** transports heat far faster than molecular conduction (eddy diffusivity $\varepsilon_H >> \alpha$).
- Because the core is so well mixed, almost the whole temperature drop is squeezed into the very thin viscous sublayer, so the wall gradient $(\partial T/\partial y)_0$ is very steep.
- Therefore $h_{turbulent}$ is several times $h_{laminar}$ (and so is wall friction). Example: flat plate, laminar $Nu \propto Re^{0.5}$, turbulent $Nu \propto Re^{0.8}$.

Dimensionless numbers and their significance (2020 4a, 2019 6a, 2023 6a, 2021 6b, 2020 6b)

Number	Definition	Physical meaning
Reynolds Re	$VL_c/\nu = \rho VL_c/\mu$	inertia forces / viscous forces; decides laminar or turbulent (plate 5×10^5, pipe 2300)
Prandtl Pr	$\nu/\alpha = \mu c_p/k$	momentum diffusivity / thermal diffusivity; relative thickness of velocity and thermal boundary layers (gases 0.7, water 1.7-13.7, oils 100-40,000, liquid metals 0.01)
Nusselt Nu	hL_c/k_{fluid}	convection heat transfer / conduction across the same fluid layer; Nu = 1 means pure conduction; dimensionless h
Grashof Gr	$g\beta(T_s - T_\infty)L_c^3/\nu^2$	buoyancy forces / viscous forces; plays the role of Re in natural convection
Rayleigh Ra	Gr·Pr	buoyancy-driven flow strength; decides laminar ($< 10^9$) or turbulent natural convection on a vertical plate
Stanton St	$h/(\rho c_p V) = Nu/(RePr)$	heat transferred / heat capacity of the flow

Grashof vs Reynolds (2020 6b): Re compares inertia to viscous forces in *forced* flow where the velocity is imposed; Gr compares buoyancy to viscous forces in *natural* convection where there is no imposed velocity (velocity is created by the temperature difference). If $Gr/Re^2 << 1$ natural convection can be ignored; $>> 1$ forced convection can be ignored; ≈ 1 mixed convection.

Film temperature (2022 6b): $T_f = (T_s + T_\infty)/2$, the arithmetic mean of the surface and free-stream temperatures. Fluid properties vary across the boundary layer, so they are evaluated at T_f as a representative average; this makes constant-property correlations ($Nu = CRe^mPr^n$) accurate for external flow.

Dimensional analysis: Nu = f(Re, Pr) (2021 6a)

Step (write this)	Why / what it means
$h = f(\rho, \mu, c_p, k, V, L)$: 7 variables	Variables affecting h in forced convection.
Basic dimensions M, L, T, θ (4) ⇒ 7 - 4 = 3 π groups (Buckingham π theorem)	Dimensions: h [$MT^{-3}\theta^{-1}$], ρ [ML^{-3}], μ [$ML^{-1}T^{-1}$], c_p [$L^2T^{-2}\theta^{-1}$], k [$MLT^{-3}\theta^{-1}$], V [LT^{-1}], L [L].
Repeating variables: ρ, V, L, k (contain M, L, T, θ)	They must together contain all four dimensions.
$\pi_1 = \rho^aV^bL^ck^dh$: θ: -d - 1 = 0 ⇒ d = -1; M: a + d + 1 = 0 ⇒ a = 0; T: -b - 3d - 3 = 0 ⇒ b = 0; L: -3a + b + c + d = 0 ⇒ c = 1 ⇒ **$\pi_1 = hL/k = Nu$**	Equate the exponents of each dimension to zero.
$\pi_2 = \rho^aV^bL^ck^d\mu$ ⇒ d = 0, a = -1, b = -1, c = -1 ⇒ $\pi_2 = \mu/(\rho VL) = 1/Re$	Same method.
$\pi_3 = \rho^aV^bL^ck^dc_p$ ⇒ d = -1, a = 1, b = 1, c = 1 ⇒ $\pi_3 = \rho Vc_pL/k = RePr$	
$Nu = f(Re, Pr)$, usually $Nu = C\ Re^mPr^n$	Since $\pi_3/\pi_2^{-1} = Pr$; C, m, n come from experiments.

Methods of finding h (2022 8b)

1. **Dimensional analysis + experiments:** gives $Nu = CRe^{m}Pr^{n}$; constants from tests. Most correlations used in design come from this.
2. **Exact analytical solution** of the boundary-layer equations (e.g. Blasius similarity solution for a flat plate).
3. **Approximate integral method** (von Kármán): assume velocity and temperature profiles, satisfy the integral momentum and energy equations.
4. **Analogy between heat and momentum transfer** (Reynolds, Colburn): $St\,Pr^{2/3} = C_f/2$, so h is found from measured friction, which is easier to measure.
5. **Numerical method** (finite difference/volume, CFD): solve the full equations on a grid by computer for complex geometries.

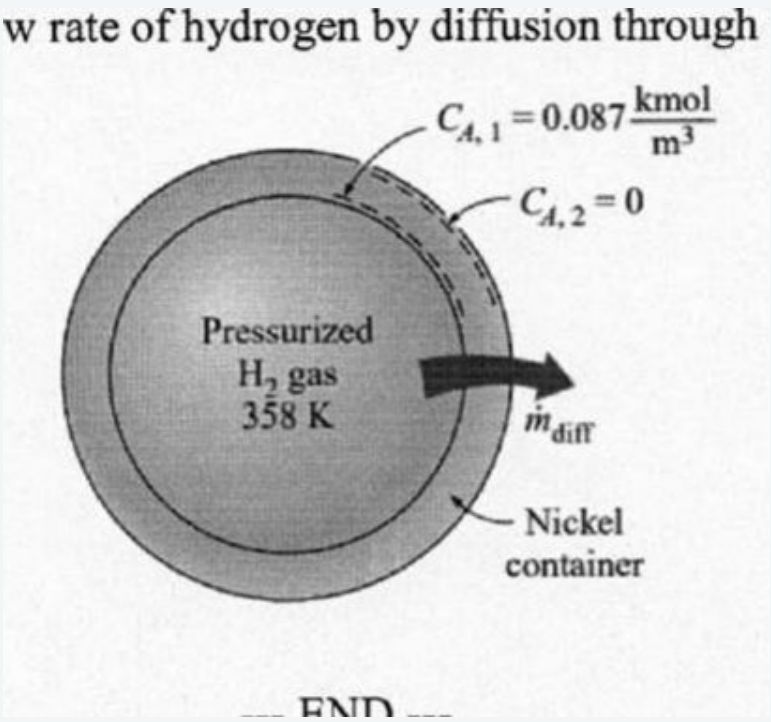
> **Memory trick:** "**Convection = conduction at the wall + carrying away**". $h \approx k/\delta_t$. Re = inertia/viscous, Pr = ν/α, Nu = hL/k, Gr = buoyancy/viscous, Ra = GrPr.

B7 Mass transfer: Fick's law, diffusion, analogies, numericals Pass set

Very likely · 100% Medium

PAST-PAPER QUESTIONS

- **2023 Q7(a)** [10] Analogy between heat and mass transfer; examples of (i) liquid-to-gas, (ii) solid-to-gas mass transfer.
- **2023 Q8(a)** [8] State Fick's law; differences between absorption and adsorption.
- **2023 Q8(b)** [17] Toluene at the bottom of a 1.524 m deep vessel, air at 1 atm, 18.7 °C, D = 0.826×10^{-5} m^2/s, p_{sat} = 0.026 atm. Evaporation rate per unit area.
- **2022 Q8(a)** [8] State Fick's law; differences between absorption and adsorption.
- **2021 Q8(b)** [10] State Fick's law. How does gas mass diffusivity change with temperature and pressure?
- **2021 Q8(a)** [10] Differences between absorption and adsorption; schematic of simple distillation.
- **2021 Q8(c)** [15] H_2 at 358 K in a nickel sphere 4.8 m OD, 6 cm thick; C_A inside 0.087 kmol/m^3, outside 0. Mass flow rate of H_2 by diffusion.



- **2020 Q6(a)** [12] Define mass diffusivity. Explain Fick's law. Discuss analogies between heat and mass transfer.
- **2020 Q6(c)** [10] Difference between mass transfer and bulk fluid flow? Can mass transfer occur in a homogeneous medium?
- **2019 Q8(a)** [6] State and explain Fick's law of diffusion.
- **2019 Q8(b)** [14] Derive the diffusion rate for steady equimolar counter-diffusion of A and B.
- **2019 Q8(c)** [15] Water diffusing from the bottom of a beaker 10 cm diameter, 20 cm long, into dry air at 32 °C; D = 0.256 cm^2/s. Diffusion rate.

UNDERSTAND **Read this to understand; you do not need to write it**

Mass transfer is the twin of heat transfer: concentration difference drives it the way temperature difference drives heat. **Fick's law** j = −D dC/dx is Fourier's law with D (mass

diffusivity) in place of α. Evaporation from a liquid at the bottom of a tube into stagnant air (Stefan tube) is faster than plain diffusion because the air must stay still while vapour flows up, giving a log term.

CALCULATOR **Evaporation through stagnant air (Stefan tube)**

Change any number; every step is recalculated and shown with the numbers put in.

1. Area: π × **0.1** × **0.1** / 4 = **0.00785** m²
2. Molar flux: N = (Dp/R_uTL) ln[(p − p_{A2})/(p − p_{A1})] = **2.560 × 10^{-5}** × **101.3** / (8.314 × **305** × **0.2**) × ln((**101.3** - **0**) / (**101.3** - **4.76**)) = **2.461 × 10^{-7}** kmol/m²s
3. Evaporation rate: **2.461 × 10^{-7}** × **0.007854** × **18** × 3600 × 1000 = **0.125** g/h

Saturation pressure of water at 32 °C = 4.76 kPa. Equimolar counter-diffusion (no log term): N = D(p_{A1} − p_{A2})/(R_uTL).

Before you start: mass transfer in plain words

Mass transfer is the movement of a species (a chemical component) from a region of high concentration to low concentration, exactly like heat moving from high to low temperature. Example: the smell of perfume spreading, sugar dissolving in still tea, water evaporating into dry air.

Concentration can be expressed as mass fraction $w_A = \rho_A/\rho$, molar concentration C_A (kmol/m^3), or mole fraction $y_A = C_A/C = P_A/P$ for ideal gases; $C = P/(R_uT)$, R_u = 8.314 kJ/kmol K.

Fick's law and mass diffusivity (2019 8a, 2020 6a, 2021 8b, 2022 8a, 2023 8a)

Fick's law: the rate of diffusion of a species A per unit area is proportional to its concentration gradient: $\mathbf{j_A = -\rho D_{AB} dw_A/dx}$ (kg/m^2s) or, on a molar basis, $\mathbf{J_A = -C D_{AB} dy_A/dx}$ (kmol/m^2s); for constant density, $J_A = -D_{AB}dC_A/dx$. The minus sign shows diffusion goes down the concentration gradient.

Mass diffusivity D_{AB} (m^2/s): the diffusion coefficient of A through B; it measures how fast a species spreads. Typical: gases ~10^{-5}, liquids ~10^{-9}, solids 10^{-10}-10^{-34} m^2/s. **Effect of T and P for gases:** $D_{AB} \propto T^{3/2}/P$: it rises with temperature (molecules move faster) and falls with pressure (more collisions, shorter free path). So $D_{AB,2} = D_{AB,1}(P_1/P_2)(T_2/T_1)^{3/2}$.

Analogy between heat and mass transfer (2020 6a, 2023 7a)

Heat transfer	Mass transfer
Driving force: temperature difference ΔT	Driving force: concentration difference ΔC
Fourier: $q = -k\, dT/dx$	Fick: $j = -\rho D_{AB}\, dw/dx$
Thermal diffusivity α	Mass diffusivity D_{AB}
Prandtl $Pr = \nu/\alpha$, Nusselt $Nu = hL/k$	Schmidt $Sc = \nu/D_{AB}$, Sherwood $Sh = h_m L/D_{AB}$
Lewis number $Le = \alpha/D_{AB} = Sc/Pr$ links the two	Correlations swap: $Nu = CRe^m Pr^n \leftrightarrow Sh = CRe^m Sc^n$

Examples (2023 7a): (i) liquid to gas: evaporation of water from a lake or a wet cloth into air, a cooling tower; (ii) solid to gas: sublimation of naphthalene balls (moth balls) or dry ice into air, drying of wood/grain.

Mass transfer vs bulk flow (2020 6c): bulk fluid flow moves the whole fluid, all species together, driven by a pressure difference (a fan or pump); it occurs even in a homogeneous mixture. Mass transfer (diffusion) moves one species *relative to* the mixture, driven by a concentration difference. **Can mass transfer occur in a homogeneous medium?** No: a homogeneous medium has the same concentration everywhere, so there is no concentration gradient and no driving force for mass transfer (bulk flow can still occur).

Absorption vs adsorption (2021 8a, 2022 8a, 2023 8a)

Absorption	Adsorption
Substance enters the **whole volume** (bulk) of the other phase	Substance sticks only on the **surface** of a solid
Bulk phenomenon; uniform throughout	Surface phenomenon; concentration higher at the surface
Rate uniform; continues until the solution is saturated	Rate fast at first, then slows as the surface fills
Examples: CO_2 absorbed in water (soda), ammonia in water (absorption refrigerator), water in a sponge	Examples: activated charcoal removing odour, silica gel drying air, gas masks

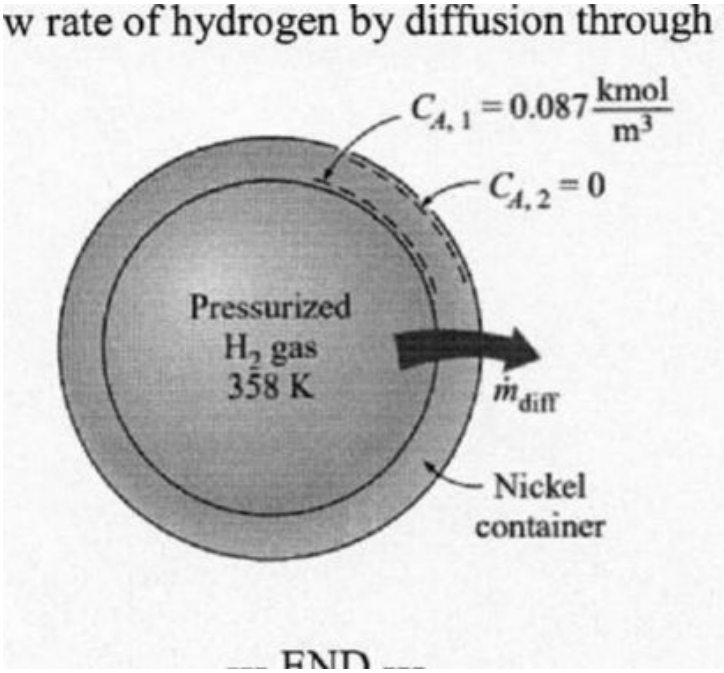
Simple distillation (2021 8a): a liquid mixture is heated in a still (flask/boiler); the more volatile component (lower boiling point) vaporises more; the vapour rises into a condenser (cooled by water) where it turns back to liquid and is collected in a receiver as the distillate, richer in the volatile component, while the less volatile component stays in the still. Sketch: heater → boiling flask with thermometer → side-arm → water-cooled condenser → receiver.

Steady equimolar counter-diffusion (2019 8b)

Step (write this)	Why / what it means
`Gases A and B in a tube of length L between two reservoirs at the same total P and T; A diffuses one way, B the other way`	Set-up: total pressure constant.
$C = C_A + C_B = P/(R_u T) = \text{constant} \Rightarrow dC_A/dx = -dC_B/dx$	Total concentration fixed.
$N_A = -N_B$ (equimolar)	Each mole of A going right is replaced by a mole of B going left, so there is no bulk motion.

$N_A = -CD_{AB}dy_A/dx$ (no bulk-flow term)	Fick's law alone.
Integrate from $x = 0$ ($y_{A,1}$) to L ($y_{A,2}$) with N_A constant (steady)	
$N_A = CD_{AB}(y_{A,1} - y_{A,2})/L = (D_{AB}/R_uT)(P_{A,1} - P_{A,2})/L$	Molar flux (kmol/m^{2}s); multiply by the area for the rate; $N_B = -N_A$. The concentration profile is linear.

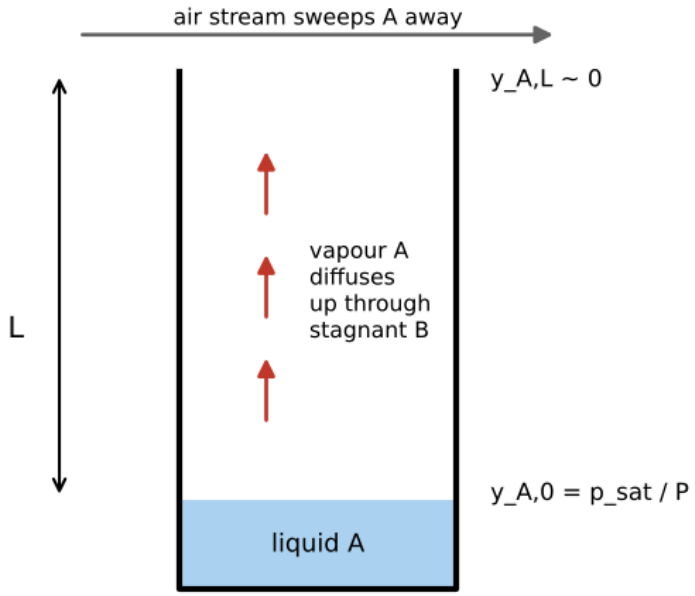
Numerical 1: hydrogen diffusing through a nickel sphere (2021 8c)



Question figure (2021 paper)

Step (write this)	Why / what it means
$r_2 = 4.8/2 = 2.4$ m, $r_1 = 2.4 - 0.06 = 2.34$ m; $D_{AB}(H_2$ in Ni, 358 K$) = 1.2\times10^{-12}$ m^2/s	Diameter as printed (4.8 m); D_{AB} from the figure/data.
Steady diffusion through a spherical shell (like heat conduction): $N = 4\pi r_1 r_2 D_{AB}(C_{A1} - C_{A2})/(r_2 - r_1)$	Same form as $Q = 4\pi k r_1 r_2 \Delta T/(r_2 - r_1)$.
$N = 4\pi \times 2.34 \times 2.4 \times 1.2\times10^{-12} \times (0.087 - 0)/0.06 = 1.23\times10^{-10}$ kmol/s	
$\dot{m} = NM = 1.23\times10^{-10} \times 2.016 = 2.48\times10^{-10}$ kg/s	Molar mass of H_2 = 2.016 kg/kmol.

Diffusion of a vapour through a stagnant gas (Stefan flow)



Stefan tube: liquid at the bottom evaporating into stagnant air

When a liquid A evaporates into a column of gas B that cannot dissolve in the liquid, B is stagnant ($N_B = 0$) and the bulk motion of the mixture helps A upward: $\mathbf{N_A = (CD_{AB}/L)\ ln[(1 - y_{A,L})/(1 - y_{A,0})]}$, with $y_{A,0} = P_{sat}/P$ at the liquid surface.

Numerical 2: evaporation of toluene (2023 8b)

Step (write this)	Why / what it means
$C = P/(R_U T) = 101.325/(8.314 \times 291.85) = 0.04176$ kmol/m^3	
$y_{A,0} = 0.026$ atm/1 atm $= 0.026$; $y_{A,L} = 0$ (fresh air at the top)	
$N_A = (0.04176 \times 0.826\times10^{-5}/1.524)\ ln[(1 - 0)/(1 - 0.026)] = 2.263\times10^{-7} \times 0.02634$	Stefan flow.
$N_A = 5.96\times10^{-9}$ kmol/m^{2}s $= 5.49\times10^{-7}$ kg/m^{2}s	Molar mass of toluene C_7H_8 = 92.14 kg/kmol.

Numerical 3: water evaporating from a beaker (2019 8c)

Step (write this)	Why / what it means
$P_{sat}(32\ °C) = 4.76$ kPa ⇒ $y_{A,0} = 4.76/101.325 = 0.0470$; $y_{A,L} = 0$ (dry air)	Steam table; total pressure assumed 1 atm.
$C = 101.325/(8.314 \times 305.15) = 0.03994$ kmol/m^3; $D_{AB} = 0.256$ cm^2/s $= 2.56\times10^{-5}$ m^2/s	Unit conversion: 1 cm^2 = 10^{-4} m^2.
$N_A = (0.03994 \times 2.56\times10^{-5}/0.20)\ ln[1/(1 - 0.0470)] = 2.46\times10^{-7}$ kmol/m^{2}s	Water vapour diffuses through stagnant air.
$A = \pi(0.05)^2 = 7.854\times10^{-3}$ m^2 ⇒ 1.93×10^{-9} kmol/s	
$\dot{m} = 1.93\times10^{-9} \times 18.015 = 3.48\times10^{-8}$ kg/s (≈ 0.125 g/h)	

Memory trick: "**Fick = Fourier with C for T and D for k**". Equimolar: linear, N = CDΔy/L. Stagnant gas: add $\ln[(1 - y_L)/(1 - y_0)]$. Absorption = sponge (bulk); adsorption = sticker (surface).

B2 Boundary-layer analysis: integral momentum and energy equations, Nu from profiles, Reynolds-Colburn analogy

Next to add · Very likely · 80% · Medium

PAST-PAPER QUESTIONS

- **2023 Q6(c)** [12] Derive the energy equation for the laminar boundary layer on a flat plate.
- **2023 Q5(c)** [13] Air at 20 °C over a plate at 160 °C with $(T - T_\infty)/(T_s - T_\infty) = e^{-ay}$, $a = 3200\ m^{-1}$. Find the surface heat flux and h.
- **2022 Q6(a)** [14] Derive the integral energy equation for the thermal boundary layer and its thickness.
- **2022 Q6(c)** [15] Rough plate with $h_x = ax^{-0.1}$: (i) ratio of average to local h at x, (ii) plot h_x and $\bar{h}_x$.
- **2022 Q7(a)** [10] Concept and application of the Reynolds-Colburn analogy; its expression.
- **2020 Q4(b)** [20] Laminar flow over a flat plate with the stated boundary conditions: determine the velocity boundary-layer thickness.
- **2020 Q5(a)** [10] For laminar flow over a flat plate, develop the average friction coefficient.
- **2019 Q5(b)** [15] With $\theta = 1.5(y/\delta_t) - 0.5(y/\delta_t)^3$ and $\delta_t = 4.53x/(Re_x^{1/2}Pr^{1/3})$, develop local Nu and average Nu from 0 to L.

Before you start: the integral method in one picture

Instead of solving the boundary-layer equations exactly, we assume a reasonable shape for the velocity (or temperature) profile, e.g. a cubic, and force it to satisfy the **integral** (whole-layer) balance of momentum or energy. This gives $\delta(x)$ or $\delta_t(x)$ and from it the friction coefficient and h.

Cubic profile $u/U = (3/2)(y/\delta) - (1/2)(y/\delta)^3$ satisfies $u(0) = 0$, $u(\delta) = U$, $\partial u/\partial y(\delta) = 0$, $\partial^2 u/\partial y^2(0) = 0$.

Velocity boundary-layer thickness (2020 4b)

Step (write this)	Why / what it means
Integral momentum equation (zero pressure gradient): $d/dx\int_0^\delta \rho u(U - u)dy = \tau_s = \mu(\partial u/\partial y)_{y=0}$	Momentum lost by the layer = wall shear (von Kármán).
BCs: $y = 0$: $u = 0$, $\partial^2 u/\partial y^2 = 0$; $y = \delta$: $u = U$, $\partial u/\partial y = 0 \Rightarrow u/U = (3/2)(y/\delta) - (1/2)(y/\delta)^3$	Fit $u = a + by + cy^2 + dy^3$ to the four conditions: $a = 0$, $c = 0$, $b = 3U/2\delta$, $d = -U/2\delta^3$.
$\int_0^\delta u(U - u)dy = (39/280)U^2\delta$	Substitute and integrate the polynomial.
$\tau_s = \mu(3U/2\delta)$	Differentiate the profile at $y = 0$.
$(39/280)\rho U^2 d\delta/dx = 3\mu U/(2\delta) \Rightarrow \delta d\delta = (140/13)(\nu/U)dx$	Substitute into the momentum equation.
$\delta^2/2 = (140/13)(\nu x/U)$ ($\delta = 0$ at $x = 0$)	Integrate.
$\delta/x = 4.64/\sqrt{Re_x}$	$\sqrt{(280/13)} = 4.64$. Blasius exact: $5.0/\sqrt{Re_x}$.

Average friction coefficient (2020 5a)

Step (write this)	Why / what it means
$\tau_s = 3\mu U/(2\delta) = (3\mu U/2)(\sqrt{Re_x}/4.64x)$	Wall shear from the profile.
$C_{f,x} = \tau_s/(\rho U^2/2) = 0.646/\sqrt{Re_x}$	Local friction coefficient (Blasius exact: 0.664).

$C_f = (1/L)\int_0^L C_{f,x}dx = (0.646/L)\sqrt{(\nu/U)}\int_0^L x^{-1/2}dx$	Average over the plate.
$C_f = 1.29/\sqrt{Re_L}$ (exact Blasius: **$1.328/\sqrt{Re_L}$**)	Average = 2 × local value at x = L, because of the $x^{-1/2}$ shape.

Energy equation of the laminar boundary layer (2023 6c)

Step (write this)	Why / what it means
Element dx × dy (unit depth) in steady 2-D flow, velocities u, v, temperature T	Control volume inside the layer.
Energy carried in by flow (net, x and y): $-\rho c_p[\partial(uT)/\partial x + \partial(vT)/\partial y]dxdy$	Enthalpy convected through the faces.
Net conduction in: $k(\partial^2T/\partial x^2 + \partial^2T/\partial y^2)dxdy$	Fourier's law on each face.
Use continuity $\partial u/\partial x + \partial v/\partial y = 0$: $\partial(uT)/\partial x + \partial(vT)/\partial y = u\partial T/\partial x + v\partial T/\partial y$	Removes the $T(\partial u/\partial x + \partial v/\partial y)$ term.
$\rho c_p(u\partial T/\partial x + v\partial T/\partial y) = k(\partial^2T/\partial x^2 + \partial^2T/\partial y^2) + \mu\Phi$	Φ = viscous dissipation.
Boundary-layer approximation: $\partial^2T/\partial x^2 \ll \partial^2T/\partial y^2$, $\Phi \approx (\partial u/\partial y)^2$	The layer is thin.
$u\partial T/\partial x + v\partial T/\partial y = \alpha\partial^2T/\partial y^2 + (\nu/c_p)(\partial u/\partial y)^2$	Dissipation is negligible at low speed: $u\partial T/\partial x + v\partial T/\partial y = \alpha\partial^2T/\partial y^2$.

Integral energy equation and thermal layer thickness (2022 6a)

Step (write this)	Why / what it means
Integrate the energy equation across the thermal layer from y = 0 to H (> δ_t)	Integral form, like the momentum one.
$d/dx\int_0^{\delta_t} u(T_\infty - T)dy = \alpha(\partial T/\partial y)_{y=0}$	Energy carried by the layer = heat conducted in at the wall.
Profiles: $\theta = (T - T_s)/(T_\infty - T_s) = (3/2)(y/\delta_t) - (1/2)(y/\delta_t)^3$, u/U cubic in y/δ	Same shape as the velocity profile.
Let $\zeta = \delta_t/\delta < 1$: integral = $U(T_\infty - T_s)\delta(3/20\zeta^2 - 3/280\zeta^4) \approx (3/20)U\Delta T\delta\zeta^2$	Neglect the small ζ^4 term.
$(3/20)U\, d(\delta\zeta^2)/dx = 3\alpha/(2\delta\zeta)$; with δ from the momentum solution	Substitute the wall gradient.
$\zeta^3 = (13/14)/Pr$ (with an unheated start length of 0) ⇒ **$\delta_t/\delta = Pr^{-1/3}/1.026$**	Solve; $\zeta \approx Pr^{-1/3}$.
$\delta_t = 4.53x/(Re_x^{1/2}Pr^{1/3})$	Combine with $\delta = 4.64x/\sqrt{Re_x}$.

Local and average Nu from the given profile (2019 5b)

Step (write this)	Why / what it means
$\theta = (3/2)(y/\delta_t) - (1/2)(y/\delta_t)^3 \Rightarrow (\partial\theta/\partial y)_0 = 3/(2\delta_t)$	Differentiate at the wall.
$h_x = k(\partial\theta/\partial y)_0 = 3k/(2\delta_t)$	$h = -k(\partial T/\partial y)_0/(T_s - T_\infty)$.
$Nu_x = h_x x/k = 1.5x/\delta_t = (1.5/4.53)Re_x^{1/2}Pr^{1/3}$	Insert δ_t.
$Nu_x = 0.332Re_x^{1/2}Pr^{1/3}$	Local.

$\bar{h} = (1/L)\int_0^L h_x dx$, $h_x \propto x^{-1/2} \Rightarrow \bar{h} = 2h_{x=L}$	
$N\bar{u} = 0.664 Re_L^{1/2} Pr^{1/3}$	Average from 0 to L.

Numerical: h from an exponential temperature profile (2023 5c)

Step (write this)	Why / what it means
$T_f = (160 + 20)/2 = 90$ °C: $k_{air} = 0.03024$ W/m K (Cengel A-15)	Fluid property at the film temperature.
$T - T_\infty = (T_s - T_\infty)e^{-ay} \Rightarrow (\partial T/\partial y)_0 = -a(T_s - T_\infty)$	Differentiate at y = 0.
$q_s = -k(\partial T/\partial y)_0 = ka(T_s - T_\infty) = 0.03024 \times 3200 \times 140 = 13{,}550$ W/m^2	Conduction into the fluid at the wall.
$h = q_s/(T_s - T_\infty) = ka = 96.8$ W/m^{2}K	

Rough plate with $h_x = ax^{-0.1}$ (2022 6c)

Step (write this)	Why / what it means
$\bar{h}_x = (1/x)\int_0^x ax^{-0.1}dx = (1/x)(a\, x^{0.9}/0.9)$	Average from the leading edge to x.
(i) $\bar{h}_x/h_x = 1/0.9 = 1.11$	Average is 11% above the local value.
(ii) Both curves fall as $x^{-0.1}$; $\bar{h}_x$ lies 11% above h_x everywhere, both infinite at x = 0	Sketch two falling curves, the average one higher.

Reynolds-Colburn analogy (2022 7a)

The momentum equation $u\partial u/\partial x + v\partial u/\partial y = v\partial^2 u/\partial y^2$ and the energy equation $u\partial T/\partial x + v\partial T/\partial y = \alpha\partial^2 T/\partial y^2$ have the same form. If Pr = 1 ($v = \alpha$) and dP/dx = 0, the dimensionless velocity and temperature profiles are identical, so wall friction and wall heat flux are linked: **Reynolds analogy** St = $C_f/2$. Colburn extended it to 0.6 < Pr < 60: **$St\, Pr^{2/3} = C_{f,x}/2$** ($j_H$ = j-factor). **Application:** measure or calculate the friction coefficient (easy) and get h without a heat test; e.g. a flat plate $C_{f,x} = 0.664/\sqrt{Re_x}$ gives $Nu_x = 0.332 Re_x^{1/2} Pr^{1/3}$.

> **Memory trick:** "**4.64, 4.53, 0.332, 0.664, 1.328**": δ, δ_t, local Nu, average Nu, average C_f. Average = 2 × local at L (laminar). $St\, Pr^{2/3} = C_f/2$.

B5 Flow across tube banks Pass set Very likely · 87% Hard

PAST-PAPER QUESTIONS

- **2023 Q7(b)** [25] Geothermal water at 120 °C in tubes; air 20 °C, 1 atm, 4.5 m/s; D = 1.5 cm in-line S_L = S_T = 5 cm, 6 rows of 10 tubes. Heat transfer per unit length and pressure drop.
- **2022 Q7(c)** [20] Water 24 → 74 °C across a staggered bank, D = 2.5 cm, S_L/D = 1.5, S_T/D = 2, tubes at 100 °C, u_∞ = 0.3 m/s. Find (i) h, (ii) number of rows.
- **2021 Q7(b)** [20] Air 325 K, 8 m/s across in-line bank, D = 1.9 cm, S_L/D = S_T/D = 2, tubes at 375 K, 15 rows, 20 tubes per row, 0.75 m. Find (i) pressure drop, (ii) exit temperature, (iii) heat transfer.
- **2019 Q7(a)** [20] Staggered bank D = 16.4 mm, S_L = 34.3 mm, S_T = 31.3 mm, 7 rows of 8 tubes, surface 70 °C, air 15 °C at 6 m/s, C_2 = 0.95, m = 0.6. Find h and heat transfer.

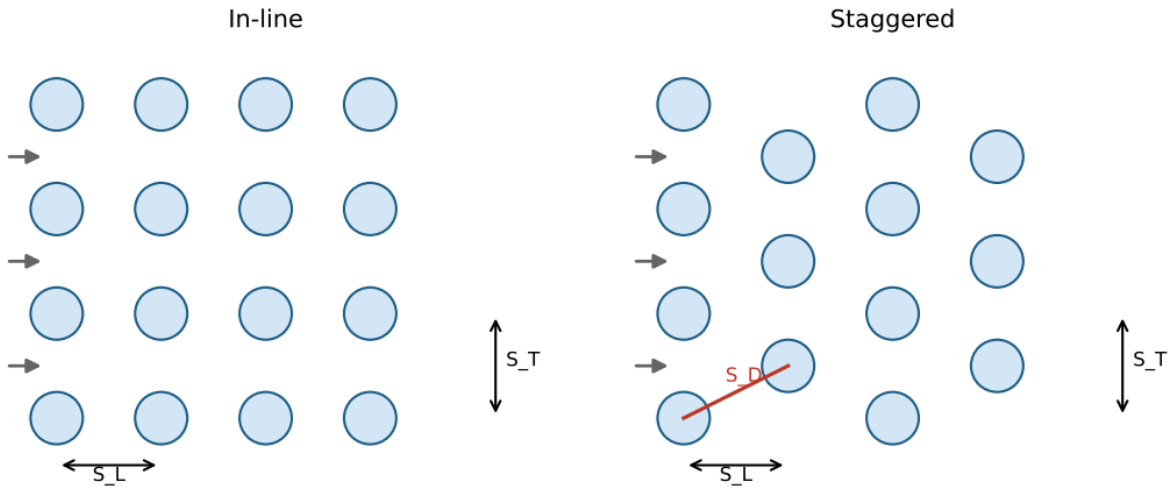
★ REMEMBER IT

- Tube bank recipe: properties at mean fluid T → V_{max} (in-line: $S_T/(S_T - D)$ V; staggered: check the diagonal) → $Re_D = V_{max}D/\nu$ → Zukauskas $Nu = C\,Re^m\,Pr^{0.36}(Pr/Pr_s)^{1/4}$ × row correction → h → exit T from $\exp(-\pi DNh/\rho V N_T S_T c_p)$ → Q with ΔT_{lm}.

Before you start: tube banks

Heat exchangers often have many tubes with a fluid flowing across them. Tubes are arranged **in-line** (one behind another) or **staggered** (shifted). S_T = transverse pitch (across the flow), S_L = longitudinal pitch (along the flow), S_D = diagonal pitch = $\sqrt{[S_L^2 + (S_T/2)^2]}$.

Recipe: V_{max} → $Re_D = V_{max}D/\nu$ → Zukauskas Nu → row correction → h → T_e from the exponential law → $Q = hA_s\Delta T_{lm}$ → $\Delta P = N_L f \chi \rho V_{max}^2/2$.



In-line and staggered tube banks with the pitches

Arrangement	Range of Re_D	Zukauskas correlation ($N_L \geq 16$; properties at T_m, Pr_s at T_s)
In-line	$1000\text{-}2\times10^5$	$Nu = 0.27Re^{0.63}Pr^{0.36}(Pr/Pr_s)^{0.25}$
Staggered	$1000\text{-}2\times10^5$	$Nu = 0.35(S_T/S_L)^{0.2}Re^{0.6}Pr^{0.36}(Pr/Pr_s)^{0.25}$
Both	fewer than 16 rows	multiply Nu by the row factor F (Cengel Table 7-3), e.g. in-line 6 rows: 0.945; 15 rows: 0.99

Numerical 1: geothermal water heating air (2023 7b, Cengel Example 7-7)

Step (write this)	Why / what it means
Mean air temperature about 60 °C (estimate): $k = 0.02808$, $v = 1.896\times10^{-5}$, $c_p = 1007$, $Pr = 0.7202$, $Pr_s(120\ °C) = 0.7073$; $\rho_{inlet}(20\ °C) = 1.204$	Exit temperature is unknown at first: guess T_m, check later.
$V_{max} = S_TV/(S_T - D) = 5 \times 4.5/(5 - 1.5) = 6.43$ m/s	Air speeds up between the tubes.
$Re_D = 6.43 \times 0.015/1.896\times10^{-5} = 5086$	
$Nu = 0.27(5086)^{0.63}(0.7202)^{0.36}(0.7202/0.7073)^{0.25} = 52.1$; 6 rows: $Nu = 0.945 \times 52.1 = 49.3$	In-line correlation + row factor.
$h = 49.3 \times 0.02808/0.015 = 92.2$ W/m^{2}K	
Per m of tube length: $A_s = 60\pi D = 2.827$ m^2; $\dot{m} = \rho V(N_TS_T) = 1.204 \times 4.5 \times 10 \times 0.05 = 2.709$ kg/s	
$T_e = 120 - 100\ exp(-2.827 \times 92.2/(2.709 \times 1007)) = 29.1$ °C	
$\Delta T_{lm} = [(120 - 20) - (120 - 29.1)]/ln(100/90.9) = 95.4$ °C	
$Q = hA_s\Delta T_{lm} = 92.2 \times 2.827 \times 95.4 = 24.9$ kW per m length	
$\Delta P = N_Lfx\rho V_{max}^2/2 = 6 \times 0.16 \times 1 \times 1.059 \times 6.43^2/2 = 21$ Pa	f = 0.16 and $\chi = 1$ from Cengel Fig. 7-27 ($S_L/D = S_T/D = 3.33$).

Numerical 2: air across an in-line bank (2021 7b)

Step (write this)	Why / what it means
$D = 0.019$ m, $S_T = S_L = 0.038$ m, $N_L = 15$, $N_T = 20$, tube length 0.75 m	Read the geometry.
Air properties at about 60 °C: $\rho = 1.060$, $k = 0.0288$, $v = 1.897\times10^{-5}$, $Pr = 0.703$, $c_p = 1008$; $Pr_s(375\ K) = 0.700$	Mean air temperature estimated, then checked.
$V_{max} = 0.038 \times 8/(0.038 - 0.019) = 16$ m/s; $Re = 16 \times 0.019/1.897\times10^{-5} = 16{,}030$	
$Nu = 0.99 \times 0.27 \times 16{,}030^{0.63} \times 0.703^{0.36} \times (0.703/0.700)^{0.25} = 106.1$; $h = 106.1 \times 0.0288/0.019 = 160.9$ W/m^{2}K	Row factor 0.99 for 15 rows.

$\dot{m} = \rho V(N_T S_T L) = 1.086 \times 8 \times (20 \times 0.038 \times 0.75) = 4.95$ kg/s (inlet density); $A_s = 300\pi \times 0.019 \times 0.75 = 13.43$ m^2	
(ii) $T_e = 375 - 50\exp(-13.43 \times 160.9/(4.95 \times 1008)) = 342.6$ K (69.4 °C)	
$\Delta T_{lm} = (50 - 32.4)/\ln(50/32.4) = 40.6$ K; **(iii) $Q = 160.9 \times 13.43 \times 40.6 = 87.7$ kW**	
(i) $\Delta P = N_L f \chi \rho V_{max}^2/2 = 15 \times 0.2 \times 1 \times 1.060 \times 16^2/2 \approx 0.41$ kPa	f ≈ 0.2 read from the in-line friction chart at Re = 1.6×10^4, $S_L/D = 2$; $\chi = 1$ for a square array.

Numerical 3: water across a staggered bank, number of rows (2022 7c)

Step (write this)	Why / what it means
$T_m = (24 + 74)/2 = 49$ °C: $k = 0.639$, $\nu = 0.562\times10^{-6}$, $Pr = 3.63$, $c_p = 4181$, $\rho(24\ °C) = 997$; $Pr_s(100\ °C) = 1.75$	
$S_T = 2D = 0.05$, $S_L = 1.5D = 0.0375$; $S_D = \sqrt{(0.0375^2 + 0.025^2)} = 0.0451 > (S_T + D)/2 = 0.0375$	So the maximum velocity is in the transverse gap.
$V_{max} = S_T V/(S_T - D) = 0.05 \times 0.3/0.025 = 0.6$ m/s; $Re = 0.6 \times 0.025/0.562\times10^{-6} = 26{,}680$	
$Nu = 0.35(2/1.5)^{0.2}(26{,}680)^{0.6}(3.63)^{0.36}(3.63/1.75)^{0.25} = 320$	Staggered correlation (many rows, so no correction).
(i) $h = 320 \times 0.639/0.025 = 8190$ W/m^{2}K	
Per unit tube length, per transverse pitch: $\dot{m}c_p = \rho V S_T c_p$; area per row = πD	Energy balance on one column of tubes.
$\ln[(T_s - T_i)/(T_s - T_e)] = N_L \pi D h/(\rho V S_T c_p) \Rightarrow N_L = \ln(76/26) \times 997 \times 0.3 \times 0.05 \times 4181/(\pi \times 0.025 \times 8190)$	
(ii) $N_L = 104$ rows	

Numerical 4: staggered bank, given C_2 and m (2019 7a, Incropera Example 7.6)

Step (write this)	Why / what it means
Air at about 15 °C: $\rho = 1.226$, $k = 0.0255$, $\nu = 1.466\times10^{-5}$, $Pr = 0.709$, $c_p = 1006$; $Pr_s(70\ °C) = 0.702$	
$S_D = \sqrt{(34.3^2 + 15.65^2)} = 37.7$ mm $> (S_T + D)/2 = 23.9$ mm $\Rightarrow V_{max} = S_T V/(S_T - D) = 31.3 \times 6/14.9 = 12.6$ m/s	
$Re = 12.6 \times 0.0164/1.466\times10^{-5} = 14{,}100$; $C_1 = 0.35(S_T/S_L)^{0.2} = 0.344$	m = 0.6 as given.
$Nu = C_2 C_1 Re^{0.6} Pr^{0.36}(Pr/Pr_s)^{0.25} = 0.95 \times 0.344 \times 14{,}100^{0.6} \times 0.709^{0.36} \times 1.003 = 89.2$	

$h = 89.2 \times 0.0255/0.0164 = 139\ W/m^2K$	Incropera's own answer is 135.6 (slightly different property values).
Per metre of tube length: $\dot{m} = \rho V(N_T S_T) = 1.226 \times 6 \times 8 \times 0.0313 = 1.84$ kg/s; $A_s = 56\pi D = 2.885\ m^2$; $T_e = 70 - 55\exp(-hA_s/\dot{m}c_p) = 25.7$ °C	
$q' = hA_s\Delta T_{lm} = 139 \times 2.885 \times 49.5 = 19.8$ kW per metre	Incropera: 19.4 kW/m.

> **Memory trick: "Squeeze (V_{max}), Reynolds, Zukauskas, rows, exp, LMTD (log mean temperature difference)"**. In-line $0.27Re^{0.63}$; staggered $0.35(S_T/S_L)^{0.2}Re^{0.6}$; both $\times Pr^{0.36}(Pr/Pr_s)^{0.25}$.

B6 Natural (free) convection Pass set Likely · 67% Medium

PAST-PAPER QUESTIONS

- **2023 Q6(b)** [15] Same 0.6 m × 0.6 m plate problem (vertical, hot up, hot down).
- **2022 Q7(b)** [5] Differences between forced and natural convection with examples.
- **2022 Q8(c)** [17] 6 m long, 8 cm horizontal hot water pipe at 70 °C in a 20 °C room: natural convection heat loss.
- **2019 Q7(b)** [15] 0.6 m × 0.6 m plate at 90 °C in a 30 °C room, other side insulated: heat transfer if (i) vertical, (ii) horizontal hot face up, (iii) hot face down.

UNDERSTAND **Read this to understand; you do not need to write it**

Hot air is lighter, so it rises along a hot plate on its own: natural convection. Its strength is measured by the **Rayleigh number Ra = gβΔTL^{3}Pr/ν²** (β = 1/T_f in kelvin for ideal gases). Then Nu = C Ra^n, h = Nu k/L, Q = hAΔT. A horizontal plate uses L = A/perimeter.

CALCULATOR **Natural convection from a plate (2019 7(b) / 2023 6(b))**

Change any number; every step is recalculated and shown with the numbers put in.

1. Film temperature: (**90** + **30**) / 2 + 273 = **333.0** K
2. Area: **0.6** × **0.6** = **0.360** m²
3. Ra, vertical (L = a): Ra = gβΔTL³Pr/ν² = 9.81 / **333** × (**90** - **30**) × pow(**0.6**,3) / (**1.896 × 10^{-5}** × **1.896 × 10^{-5}**) × **0.7202** = **7.649 × 10^8**
4. Q vertical, Nu = 0.59$Ra^{1/4}$: 0.59 × pow(**7.649 × 10^8**,0.25) × **0.02808** / **0.6** × **0.36** × (**90** - **30**) = **99.2** W
5. Horizontal L = A/perimeter: **0.6** / 4 = **0.150** m
6. Ra horizontal: **7.649 × 10^8** × pow(**0.15** / **0.6**,3) = **1.195 × 10^7**
7. Q hot face up, Nu = 0.54$Ra^{1/4}$: 0.54 × pow(**1.195 × 10^7**,0.25) × **0.02808** / **0.15** × **0.36** × (**90** - **30**) = **128.4** W
8. Q hot face down, Nu = 0.27$Ra^{1/4}$: 0.27 × pow(**1.195 × 10^7**,0.25) × **0.02808** / **0.15** × **0.36** × (**90** - **30**) = **64.2** W

Before you start: natural convection

A hot surface heats the fluid next to it; the warm fluid is lighter and rises, and cooler fluid takes its place. No fan is needed. The driving force is buoyancy, measured by the Grashof number Gr = gβ(T_s − T_∞)L_c^3/ν^2 and Ra = GrPr. For gases β = 1/T_f (in kelvin).

Recipe: T_f → properties, β → Ra → correct correlation for the geometry → h = Nu k/L_c → Q = hAΔT.

Forced vs natural convection (2022 7b)

Forced convection	Natural (free) convection
Fluid motion created by an external device: fan, pump, wind	Fluid motion created by buoyancy from density differences due to heating
Velocity known; Nu = f(Re, Pr)	Velocity unknown (results from ΔT); Nu = f(Gr, Pr) = f(Ra)
Higher h (25-250 W/m^2K for gases)	Lower h (2-25 W/m^2K for gases)
Examples: car radiator with fan, air conditioner coil, cooling a CPU with a fan	Examples: room radiator, electric transformer in air, hot tea cooling, hot-water pipe in a room

Numerical 1: hot 0.6 m × 0.6 m plate in three positions (2019 7b, 2023 6b)

Step (write this)	Why / what it means
$T_f = (90 + 30)/2 = 60$ °C: $k = 0.02808$, $\nu = 1.896\times10^{-5}$, $Pr = 0.7202$, $\beta = 1/333\ K^{-1}$; $A = 0.36\ m^2$, $\Delta T = 60$ °C	Air properties at the film temperature.
(i) Vertical, $L_c = 0.6$ m: $Ra = 9.81(1/333)(60)(0.6)^3/(1.896\times10^{-5})^2 \times 0.7202 = 7.65\times10^8$	Laminar ($< 10^9$).
$Nu = 0.59Ra^{1/4} = 98.1$; $h = 98.1 \times 0.02808/0.6 = 4.59\ W/m^2K$; **$Q = 4.59 \times 0.36 \times 60 = 99$ W**	Churchill-Chu (Cengel's choice) gives Nu = 113 and Q = 115 W; either is accepted if stated.
(ii)/(iii) Horizontal: $L_c = A/p = 0.36/2.4 = 0.15$ m; $Ra = 7.65\times10^8 \times (0.15/0.6)^3 = 1.195\times10^7$	$Ra \propto L^3$.
(ii) Hot surface facing up: $Nu = 0.54Ra^{1/4} = 31.7$, $h = 5.94\ W/m^2K$, **$Q = 128$ W**	Ra is just above 10^7; $0.15Ra^{1/3}$ gives 139 W (Cengel's answer).
(iii) Hot surface facing down: $Nu = 0.27Ra^{1/4} = 15.9$, $h = 2.97\ W/m^2K$, **$Q = 64$ W**	Warm air is trapped under the plate: the poorest case.

Ranking: hot face up > vertical > hot face down: warm air rises freely from an upward-facing hot plate but must creep around the edges of a downward-facing one.

Numerical 2: hot-water pipe in a room (2022 8c)

Step (write this)	Why / what it means
$T_f = (70 + 20)/2 = 45$ °C: $k = 0.02699$, $\nu = 1.750\times10^{-5}$, $Pr = 0.7241$, $\beta = 1/318\ K^{-1}$; $L_c = D = 0.08$ m	
$Ra = 9.81(1/318)(50)(0.08)^3/(1.750\times10^{-5})^2 \times 0.7241 = 1.87\times10^6$	
$Nu = \{\{0.6 + 0.387Ra^{1/6}/[1 + (0.559/Pr)^{9/16}]^{8/27}\}\}^2 = 17.4$	Churchill-Chu for a horizontal cylinder.
$h = 17.4 \times 0.02699/0.08 = 5.87\ W/m^2K$; $A = \pi DL = \pi \times 0.08 \times 6 = 1.508\ m^2$	
$Q = 5.87 \times 1.508 \times 50 = 442$ W	Cengel Example 9-1 (in Sohag sir's notes) gives 443 W; small differences come from property tables.

Memory trick: "Up 0.54, side 0.59, down 0.27" (all $Ra^{1/4}$); horizontal L_c = A/p, vertical L_c = height, cylinder L_c = D. $\beta = 1/T_f(K)$.

WRITE IN EXAM **Write this in the exam (this much is enough for full marks)**

2019 7(b) 0.6 m × 0.6 m plate, 90 °C, room 30 °C. T_f = 60 °C: k = 0.02808, $\nu = 1.896\times10^{-5}$, Pr = 0.7202, $\beta = 1/333\ K^{-1}$

(i) Vertical: $Ra = g\beta\Delta TL^3Pr/\nu^2$

$= 9.81(1/333)(60)(0.6)^3(0.7202)/(1.896\times10^{-5})^2$

$= \mathbf{7.65 \times 10^8}$

$Nu = 0.59Ra^{1/4} = 98.1$; $h = 98.1 \times 0.02808/0.6 = 4.59\ W/m^2K$; **Q = 4.59 × 0.36 × 60 = 99 W**

(ii) Horizontal, L = A/p = 0.15 m: $Ra = 1.195 \times 10^7$; hot up $Nu = 0.54Ra^{1/4} = 31.7$, h = 5.94, **Q = 128 W**

(iii) Hot down $Nu = 0.27Ra^{1/4} = 15.9$, h = 2.97, **Q = 64 W**

B4 Internal forced convection: fully developed flow, entry lengths, analogy, pipe numericals

Next to add · Likely · 60% · Medium

PAST-PAPER QUESTIONS

- **2023 Q8(c)** [10] "For thermally fully developed flow the axial temperature gradient is independent of the radial direction": justify.
- **2021 Q5(b)** [12] For forced convection inside a tube, prove St = f/8.
- **2021 Q5(c)** [5] How does surface roughness affect pressure drop in turbulent tube flow? And in laminar flow?
- **2021 Q5(d)** [5] Sketch hydrodynamic and thermal entry lengths for Pr = 0.7 and Pr = 7.
- **2021 Q7(a)** [15] Prove that the axial temperature gradient does not depend on radial position for uniform surface heat flux in thermally fully developed flow.
- **2021 Q6(c)** [15] Annulus 35 mm ID, 50 mm OD; water 20 → 34 °C at 2 m/s, inner surface 60 °C. Estimate h (Nu = $0.023Re^{0.8}Pr^{n}$).
- **2019 Q6(c)** [17] Oil in a 30 cm pipeline at 2 m/s, 200 m through icy lake (surface 0 °C): (i) exit temperature, (ii) heat transfer rate, (iii) pumping power.

★ REMEMBER IT

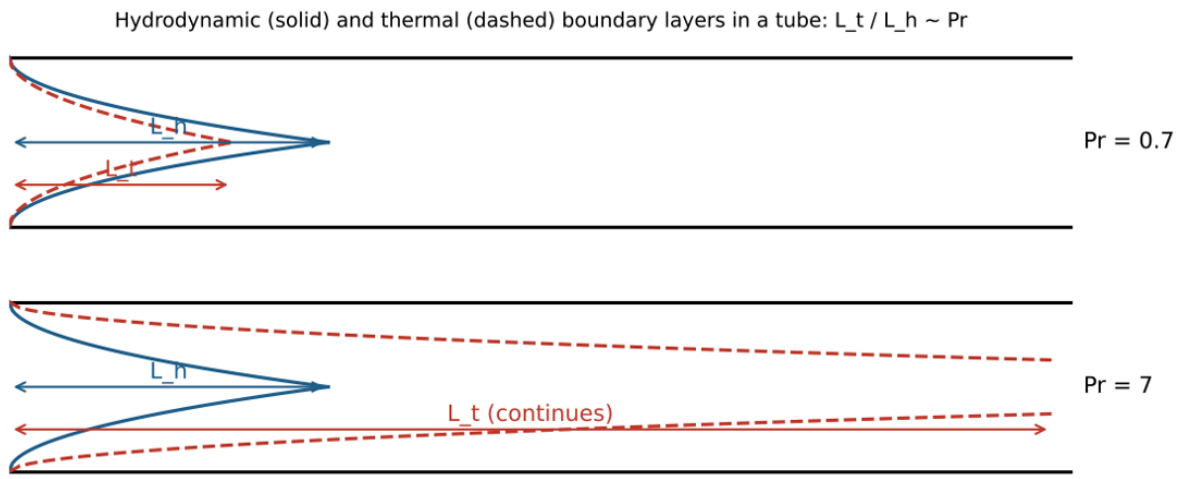
- Inside tubes: laminar if Re < 2300. Fully developed laminar, constant heat flux: Nu = 4.36; constant wall temperature: Nu = 3.66. Turbulent: Dittus-Boelter Nu = $0.023Re^{0.8}Pr^{n}$ (n = 0.4 heating, 0.3 cooling). Exit temperature: $(T_s - T_e)/(T_s - T_i) = \exp(-hA_s/\dot{m}c_p)$.

Before you start: flow inside a pipe

In a pipe the fluid is enclosed, so its temperature changes along the pipe. We use the **mean (bulk) temperature** T_m, and the energy balance $Q = \dot{m}c_p(T_e - T_i)$. Laminar if $Re = VD/\nu < 2300$; turbulent above about 10,000.

For constant surface temperature: $T_e = T_s - (T_s - T_i)\exp(-hA_s/\dot{m}c_p)$, $Q = hA_s\Delta T_{lm}$, $\Delta T_{lm} = (\Delta T_i - \Delta T_e)/\ln(\Delta T_i/\Delta T_e)$.

Entry lengths for Pr = 0.7 and Pr = 7 (2021 5d)



Hydrodynamic and thermal entry lengths in a tube

Laminar: $L_h \approx 0.05 ReD$, $L_t \approx 0.05 RePrD = Pr\, L_h$. For **Pr = 0.7** (air) the thermal layer grows slightly faster: $L_t = 0.7 L_h$. For **Pr = 7** (water) $L_t = 7 L_h$: the velocity profile becomes fully developed long before the temperature profile.

Axial temperature gradient in fully developed flow with uniform q_s (2021 7a, 2023 8c)

Step (write this)	Why / what it means
Thermally fully developed: $\partial/\partial x[(T_s - T)/(T_s - T_m)] = 0$	The shape of the dimensionless profile no longer changes with x.
$\Rightarrow (T_s - T)/(T_s - T_m) = f(r)$ only; differentiating at the wall, $h = q_s/(T_s - T_m) =$ constant	h does not change with x.
Uniform q_s and constant $h \Rightarrow T_s - T_m = q_s/h =$ constant $\Rightarrow dT_s/dx = dT_m/dx$	Wall and mean temperatures rise in parallel.
Expand: $dT_s/dx - \partial T/\partial x - f(r)(dT_s/dx - dT_m/dx) = 0$	Derivative of the product.
$\partial T/\partial x = dT_s/dx = dT_m/dx = q_s p/(\dot{m} c_p)$ = constant	The axial gradient is the same at every r (independent of radial position), from the energy balance $\dot{m} c_p dT_m = q_s p\, dx$.

Reynolds analogy for a tube: St = f/8 (2021 5b)

Step (write this)	Why / what it means
Wall shear in a tube: $\tau_s = (f/8)\rho V^2$	Definition of Darcy friction factor f.
Reynolds analogy (Pr = 1): momentum and heat are carried by the same eddies, so $q_s/\tau_s = c_p(T_s - T_m)/V$	Ratio of heat flux to momentum flux equals the ratio of the driving potentials.
$h(T_s - T_m) = (f/8)\rho V^2 c_p(T_s - T_m)/V$	Substitute $q_s = h\Delta T$.
$St = h/(\rho c_p V) = f/8$	Colburn form for Pr ≠ 1: $St\, Pr^{2/3} = f/8$.

Roughness (2021 5c): in **turbulent** flow roughness elements poke through the thin viscous sublayer, create extra eddies and raise the friction factor (Moody chart, Colebrook equation), so the pressure drop increases (h also rises somewhat). In **laminar** flow f = 64/Re regardless of roughness: the thick viscous layer covers the roughness, so it has no noticeable effect on the pressure drop.

Numerical 1: water in an annulus (2021 6c)

Step (write this)	Why / what it means
$D_h = D_o - D_i = 0.050 - 0.035 = 0.015$ m	Hydraulic diameter 4A/p of an annulus.
$T_m = (20 + 34)/2 = 27$ °C: $\nu = 0.854\times10^{-6}$ m²/s, $k = 0.610$ W/m K, $Pr = 5.84$	Water properties at the mean bulk temperature.
$Re = VD_h/\nu = 2 \times 0.015/0.854\times10^{-6} = 35{,}130 \Rightarrow$ turbulent	
$Nu = 0.023 Re^{0.8} Pr^{0.4} = 0.023 \times 35{,}130^{0.8} \times 5.84^{0.4} = 201.7$	n = 0.4 because the water is being heated.
$h = Nu\, k/D_h = 201.7 \times 0.610/0.015 = 8200$ W/m²K	

Numerical 2: oil pipeline through an icy lake (2019 6c)

Assumption: the oil inlet temperature is not given in the question; it is taken as **20 °C**, as in Cengel Example 8-6 on which the question is based. Properties of oil at 20 °C: ρ = 888.1 kg/m^3, k = 0.145 W/m K, ν = 9.429×10^{-4} m^2/s, c_p = 1880 J/kg K, Pr = 10,863.

Step (write this)	Why / what it means
$Re = VD/\nu = 2 \times 0.3/9.429\times10^{-4} = 636 \Rightarrow$ laminar	
$L_t = 0.05RePrD = 0.05 \times 636 \times 10{,}863 \times 0.3 = 103{,}700$ m >> 200 m	The whole pipe is a thermal entry region.
$Nu = 3.66 + 0.065(D/L)RePr/[1 + 0.04((D/L)RePr)^{2/3}]$ with $(D/L)RePr = 10{,}360 \Rightarrow Nu = 37.3$	Developing laminar flow correlation.
$h = 37.3 \times 0.145/0.3 = 18.0$ W/m^{2}K; $A_s = \pi DL = 188.5$ m^2; $\dot{m} = \rho(\pi D^2/4)V = 125.6$ kg/s	
(i) $T_e = 0 - (0 - 20)\exp(-18.0 \times 188.5/(125.6 \times 1880)) = 19.71$ °C	Oil barely cools: huge mass flow.
$\Delta T_{lm} = (20 - 19.71)/\ln(20/19.71) = 19.86$ °C; **(ii) $Q = hA_s\Delta T_{lm} = 18.0 \times 188.5 \times 19.86 = 67.5$ kW**	
$f = 64/Re = 0.1006$; $\Delta P = f(L/D)\rho V^2/2 = 0.1006 \times (200/0.3) \times 888.1 \times 2^2/2 = 119.1$ kPa	
(iii) $\dot{W}_{pump} = \dot{V}\Delta P = (\dot{m}/\rho)\Delta P = 0.1414 \times 119.1 = 16.8$ kW	

Memory trick: "Laminar 64/Re and 0.05Re(Pr)D; turbulent 0.023Re$^{0.8}$Pr$^{0.4\text{ heat / }0.3\text{ cool}}$". Fully developed + uniform q: everything rises at the same rate dT_m/dx.

B3 External forced convection over a flat plate: numericals Pass set

Possible · 47% Medium

PAST-PAPER QUESTIONS

- **2022 Q5(c)** [12] Air at 5 m/s, 1 atm, 28 °C over a plate at 95 °C heated from x = 12 cm: total heat transfer up to x = 40 cm.
- **2020 Q5(b)** [20] Plastic sheet 1.2 m wide, 2 mm thick, 15 m/min, 90 °C, air 3 m/s at 30 °C on both sides, 2 s in the cooling section, ε = 0.9. Find the heat transfer rate.
- **2019 Q5(c)** [15] Air at 6 kN/m^2, 300 °C, 10 m/s over a 0.5 m plate kept at 27 °C: cooling rate per unit width.

CALCULATOR **Flat plate, forced convection (laminar or mixed)**

Change any number; every step is recalculated and shown with the numbers put in.

1. Reynolds number: Re = VL/ν = **5** × **0.4** / **1.750 × 10^{-5}** = **114286**
2. Average Nusselt number: laminar 0.664Re$^{0.5}$Pr$^{1/3}$; mixed (0.037Re$^{0.8}$ − 871)Pr$^{1/3}$ = **114300** < 5×10^5 ? 0.664 × √(**114300**) × pow(**0.7**, 1 / 3) : (0.037 × pow(**114300**, 0.8) - 871) × pow(**0.7**, 1 / 3) = **199.3**
3. Average h: h = Nu k/L = **199.3** × **0.0272** / **0.4** = **13.55** W/m²K
4. Heat transfer: Q = hA(T$_s$ − T$_\infty$) = **13.55** × **0.4** × **1** × (**95** - **28**) = **363.2** W

Get ν, k, Pr from the air table at T$_f$. Unheated starting length problems (2022 5(c)): do the plate up to the end and subtract the unheated part using local h.

Before you start: flat-plate recipe

1) Film temperature T$_f$ = (T$_s$ + T$_\infty$)/2 → properties k, ν, Pr. 2) Re$_L$ = VL/ν (L in the flow direction). 3) Laminar if Re$_L$ < 5×10^5: Nu = 0.664Re$_L$$^{0.5}Pr^{1/3}$; turbulent: Nu = 0.037Re$_L$$^{0.8}Pr^{1/3}$; mixed: (0.037Re$^{0.8}$ − 871)Pr$^{1/3}$. 4) h = Nu k/L. 5) Q = hA(T$_s$ − T$_\infty$).

Numerical 1: cooling of a plastic sheet (2020 5b)

Step (write this)	Why / what it means
Sheet moves 15 m/min = 0.25 m/s; in 2 s it travels 0.5 m ⇒ cooling section is 0.5 m long and 1.2 m wide	Find the length of sheet inside the section.
Air flows across the 1.2 m width: L = 1.2 m; area of both sides A = 2 × 1.2 × 0.5 = 1.2 m^2	Flow direction is along the width (Cengel Problem 7-23).
T$_f$ = (90 + 30)/2 = 60 °C: k = 0.02808, ν = 1.896×10^{-5}, Pr = 0.7202	Air properties at 1 atm.
Re$_L$ = 3 × 1.2/1.896×10^{-5} = 1.90×10^5 < 5×10^5 ⇒ laminar	
Nu = 0.664(1.90×10^5)$^{0.5}$(0.7202)$^{1/3}$ = 259.3; h = 259.3 × 0.02808/1.2 = 6.07 W/m^{2}K	

$Q_{conv} = 6.07 \times 1.2 \times (90 - 30) = 437$ W	
$Q_{rad} = \varepsilon\sigma A(T_s^4 - T_{surr}^4) = 0.9 \times 5.67\times10^{-8} \times 1.2 \times (363^4 - 303^4) = 548$ W	Surroundings taken at the air temperature 30 °C.
$Q_{total} = 437 + 548 = 985$ W	Radiation is larger than convection at such low h.

Numerical 2: air at low pressure (2019 5c)

Step (write this)	Why / what it means
$T_f = (300 + 27)/2 = 163.5$ °C; P = 6 kPa	Low pressure changes the density only.
At 163.5 °C: $\mu = 2.457\times10^{-5}$, k = 0.03587, Pr = 0.698; $\rho = P/RT = 6000/(287 \times 436.65) = 0.0479$ kg/m^3	μ, k, Pr are nearly independent of pressure; $\rho \propto P$.
$\nu = \mu/\rho = 5.13\times10^{-4}$ m^2/s (about 17 times the 1 atm value)	
$Re_L = 10 \times 0.5/5.13\times10^{-4} = 9740 \Rightarrow$ laminar	
$Nu = 0.664 \times 9740^{0.5} \times 0.698^{1/3} = 58.1$; $h = 58.1 \times 0.03587/0.5 = 4.17$ W/m^{2}K	
Q per m width = $hL(T_\infty - T_s) = 4.17 \times 0.5 \times 273 = 569$ W/m	Heat removed from the air by the cooler plate.

Numerical 3: plate heated after an unheated starting length (2022 5c)

Side note: Interpretation: the plate is at the air temperature for $0 \le x < \xi = 0.12$ m and at 95 °C from x = 0.12 m to 0.40 m; result per metre width.

Step (write this)	Why / what it means
$T_f = (95 + 28)/2 = 61.5$ °C: k = 0.02891, $\nu = 1.912\times10^{-5}$, Pr = 0.703	
$Re_L = 5 \times 0.40/1.912\times10^{-5} = 1.046\times10^5 \Rightarrow$ laminar	
Local Nu at L with starting length: $Nu_x = 0.332Re_x^{0.5}Pr^{1/3}/[1 - (\xi/x)^{3/4}]^{1/3} = 95.5/0.5946^{1/3} = 113.5$	$(\xi/L)^{3/4} = 0.3^{0.75} = 0.405$.
$h_L = 113.5 \times 0.02891/0.40 = 8.20$ W/m^{2}K	
Average over the heated part: $\bar{h} = 2[1 - (\xi/L)^{3/4}]/(1 - \xi/L) \times h_L = 2 \times 0.5946/0.7 \times 8.20 = 13.9$ W/m^{2}K	Cengel Eq. 7-26.
$Q = \bar{h}(L - \xi)(T_s - T_\infty) = 13.9 \times 0.28 \times 67 = 262$ W per m width	Without the starting-length correction one would get 167 W: the heated part sees a thin new thermal layer, so h is higher.

Memory trick: "Film, Re, check 5×10^5, 0.664, h, Q". Low pressure: only ρ (so ν) changes. Starting length: $\times[1 - (\xi/x)^{3/4}]^{-1/3}$.